

LNG Piping System Design

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DESIGN REQUIREMENTS

1. Piping system shall be based on ASME B31.3 Process Piping Code,
2. The piping system shall be designed to withstand the design pressure,
3. The piping system shall be properly supported so that large deflection / sagging and high bending stress is prevented,
4. The pressure drop of the system shall be lower than the total dynamic head of the Cargo Pump in the FSRU,
5. The piping shall have flexibility and the stresses shall satisfy the code allowable stresses,
6. Design and size the LNG Buffer Tank using ASME BPVC Sec VIII Div.1, Design by Rule

DESIGN ACCEPTANCE CRITERIA

- Piping wall thickness that is designed based on **Internal Design Pressure** shall meet the following criteria; *(ref. ASME B31.3)*
 - For $t < D_o/6$
 - $$t = \frac{PD}{2(SEW + PY)}$$
- Piping wall thickness that is designed based on **External Design Pressure (Vacuum Pressure)** shall meet the following; *(ref. ASME BPVC Sec VIII, Div 1, UG-27)*
 - For $D_o/t > 10$
 - The calculated maximum allowable external design pressure shall be greater than the design external pressure of 0.10 Mpa or 1 barg.
 - $P_a \geq P_{vacuum}$

DESIGN ACCEPTANCE CRITERIA

- The piping span shall be the lower of the following;
 - Piping Support Span Calculation based on allowed bending stress.
 - Bending stress of the piping system shall be half of the basic allowable stress of the piping. *(ref. M.W. Kellogg Company)*

$$\text{➤ } L1 = \sqrt{\frac{10ZS}{W}}$$

- Piping Support Span Calculation Based on Allowed Deflection.
 - The allowed piping deflection or sagging between two supports is 12 mm. *(based on practice and engineering judgment)*

$$\text{➤ } L2 = \sqrt[4]{\frac{128EI\Delta}{W}}$$

DESIGN ACCEPTANCE CRITERIA

- Hydraulic Calculation Criteria
 - The Pressure Drop of the piping system shall be less than the pump's total dynamic head of **0.86 Mpa or 190 m** at 740 m³/h minimum flowrate. *(ref. Cargo Pump specification)*
 - The flow velocity of LNG within the piping system shall not be greater than **5 m/s**. *(ref. Project requirement)*
 - The calculation of major losses in pressure drop calculation shall be based on Darcy – Weisbach equation.

$$H_f = f \frac{L_e V^2}{2D_h g_c}$$

DESIGN ACCEPTANCE CRITERIA

- Piping Stress and Flexibility Analysis shall comply with the following code stresses: (*ASME B31.3*)
 - **Primary Stress (Sustained Stress)**
 - $S_L \leq S_H W$
 - **Occasional Stress**
 - $S_{LOCC} \leq 1.33 S_H$
 - **Secondary Stress (Expansion Stress)**
 - $S_E \leq f(1.25S_C + 0.25S_H)$
 - $S_E \leq f(1.25S_C + 1.25S_H - S_L) \rightarrow$ with sustained load margin
 - **Flexibility Analysis (*from the book of Peng or Sam Kannapan*)**
 - $L = \sqrt{\frac{3E}{S}} \Delta$

DESIGN ACCEPTANCE CRITERIA

- The LNG Buffer Tank shall be designed to carry and hold the LNG all the time based on the largest of the following; *ASME BPVC Section VIII, Div. 1*

- Wall thickness at Circumferential stress (Longitudinal joints), and when thickness does not exceed one-half of the inside radius or P does not exceed 0.385SE.

- $t_1 = \frac{PR}{SE - 0.6P}$

- Wall thickness at Longitudinal stress (Circumferential joint) when thickness does not exceed one-half of the inside radius or P does not exceed 1.25SE. *ASME BPVC Section VIII, Div. 1*

- $t_2 = \frac{PR}{2SE + 0.4p}$

DESIGN ACCEPTANCE CRITERIA

- The minimum thickness of ellipsoidal head.
 - The minimum required thickness at the thinnest point after forming of ellipsoidal head;
 - $t_e = \frac{PDK}{2SE-0.2P}$
 - $k = \frac{1}{6} \left[2 + \left(\frac{D}{2h} \right)^2 \right]$
- The design of the saddle support is based on Pressure Vessel Handbook by Eugene F. Megyesy.

PIPING WALL THICKNESS CALCULATION BASED ON INTERNAL DESIGN PRESSURE

Pipe Material = A358-304/304L, [Appendices R1.docx](#)

Electric-Fusion-Welded, Austenitic Chromium Nickel SS

- Yield Strength = 30 ksi = 206.84 Mpa
- Tensile Strength = 74 ksi = 517.10 Mpa

Allowable Stress is the lower of the following;

$$S_{\text{allow}} = S_y / 1.5 = 206.84 \text{ Mpa} / 1.5 = 137.9 \text{ Mpa}$$

$$S_{\text{allow}} = S_u / 3.0 = 517.10 \text{ Mpa} / 3.0 = 172.4 \text{ Mpa}$$

$$S_{\text{allow}} = \mathbf{137.9 \text{ Mpa}}$$

PIPING WALL THICKNESS CALCULATION BASED ON INTERNAL DESIGN PRESSURE

Wall Thickness Calculation for 40"

Outside Diameter = 1016 mm

Design Pressure = 1.85 Mpa

Welding Quality Factor, $E = 1.0$ (*ASME B31.3 Table A-1B, page 350*)

Weld Joint Reduction Factor, $W = 1.0$ (*Table 302.3.5, page 21*)

Coefficient $Y = 0.4$ (*ASME B31.1 Table 304.1.1, page 23*)

Mill Tolerance = 0.3 mm (*for ASTM 358, ERW and EFW*)

Corrosion Allowance = 0 mm

PIPING WALL THICKNESS CALCULATION BASED ON INTERNAL DESIGN PRESSURE

Wall Thickness for 40"

$$t = \frac{PD_o}{2(SEW + PY)}$$

$$t = \frac{1.85 \text{ Mpa} \times 1016 \text{ mm}}{2[(137.9 \text{ Mpa} \times 1 \times 1) + (1.85 \text{ Mpa} \times 0.4)]} = 6.78 \text{ mm}$$

- Design thickness, $t = 6.78 \text{ mm}$
 - $t < D_o/6$
 - $6.78 < 1016 \text{ mm} / 6 = 169.3 \rightarrow \text{"valid"}$
- Minimum required thickness, $t_m = t + c$
 $t_m = 6.78 + 0 = 6.78 \text{ mm}$

PIPING WALL THICKNESS CALCULATION BASED ON INTERNAL DESIGN PRESSURE

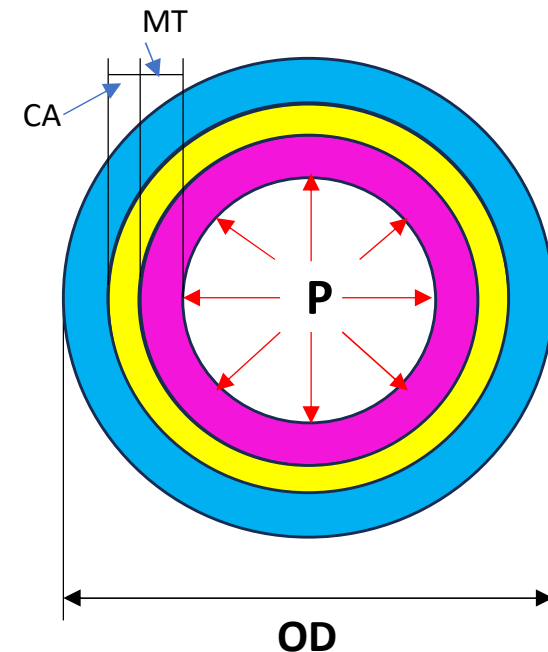
Wall Thickness for 40"

Nominal Wall Thickness, T

$$T = tm + \text{Mill Tolerance}$$

$$T = 6.78 \text{ mm} + 0.30 \text{ mm}$$

$$T = 7.08 \text{ mm}$$



Select a wall thickness that is higher than the calculated thickness T ,
The available size from the vendor is 8.74 mm thickness.

PIPING WALL THICKNESS CALCULATION BASED ON INTERNAL DESIGN PRESSURE

NPS	OD [mm]	Pipe Material	Pressure Design Thickness, t [mm]	Minimum Required Thickness $t_m = t + CA$ [mm]	Nominal Thickness, $T = t_m + MT$ [mm]	Selected Nominal thickness [mm]	SCH No.	Validity check, $t < D/6$
40	1016	A358-304/304L	6.78	6.78	7.08	8.74		169.33
28	711	A358-304/304L	4.74	4.74	5.04	6.35	SCH 10S	118.50
24	610	A358-304/304L	4.07	4.07	4.37	6.35	SCH 10S	101.67
22	559	A358-304/304L	3.73	3.73	4.03	5.54	SCH 10S	93.17
20	508	A358-304/304L	3.39	3.39	3.69	5.54	SCH 10S	84.67
18	457	A358-304/304L	3.05	3.05	3.35	4.78	SCH 10S	76.17
16	406	A358-304/304L	2.71	2.71	3.01	4.76	SCH 10S	67.67
14	356	A358-304/304L	2.38	2.38	2.68	4.78	SCH 10S	59.33
12	324	A358-304/304L	2.16	2.16	2.46	4.57	SCH 10S	54.00
10	273	A358-304/304L	1.82	1.82	2.12	4.19	SCH 10S	45.50
8	219	A358-304/304L	1.46	1.46	1.76	3.76	SCH 10S	36.50

PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

Verify Wall Thickness for 40" Pipe

Outside Diameter = 1016 mm

Nominal Thickness = 8.74 mm

Mill Tolerance = 0.30 mm

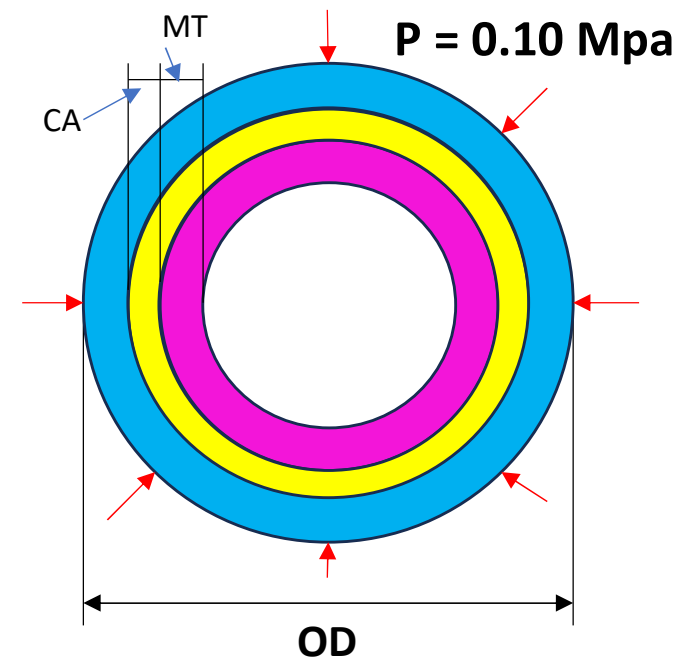
Corrosion allowance = 0 mm

Effective Length = 18,000 mm

Design Temp = 80 deg. C / -172 deg. C

Modulus of Elasticity $E = 1.95E+5$ Mpa

Design External Pressure (Vacuum) = 0.10 Mpa



PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

Verify Wall Thickness for 40" Pipe

Calculate the Minimum Thickness:

$$t = 8.74 - MT - CA$$

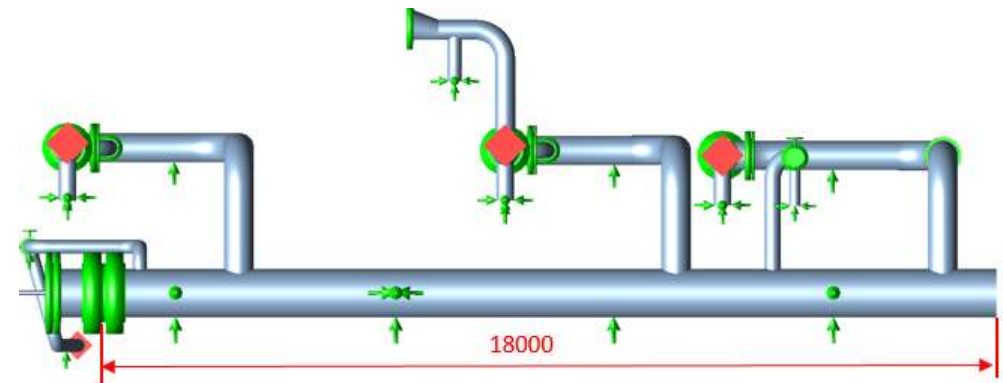
$$t = 8.74 - 0.30 - 0$$

$$t = 8.44 \text{ mm}$$

The procedure in UG-28 is valid only for $Do/t > 10$.

Calculate Do/t and L/Do

- $Do/t = 1016 \text{ mm} / 8.44 \text{ mm} = \underline{120.37 \text{ mm}} \rightarrow \text{"valid"}$
- $L/Do = 18,000 \text{ mm} / 1016 \text{ mm} = \underline{17.72 \text{ mm}}$



Obtain the value of Factor A from the Figure G or Table G.

- Figure G**

D_o/t	L/D_o	A	D_o/t	L/D_o	A	D_o/t	L/D_o	A	D_o/t	L/D_o	A	D_o/t	L/D_o	A	D_o/t	L/D_o	A
4	2.2	0.959 -01	8	5.0	0.184 -01	20	3.0	0.446 -02	30	50.0	0.122 -02	60	0.074	0.954 -01	100	0.20	0.831 -02
	2.6	0.884		7.0	0.179		3.4	0.388		30.0	0.122 -		0.10	0.556		0.40	0.364
	3.0	0.839		10.0	0.176		4.0	0.342					0.14	0.323		0.5	0.283
	4.0	0.783		20.0	0.174		5.0	0.308	40	0.12	0.864 -01		0.2	0.193		0.8	0.170
	5.0	0.759		50.0	0.174		7.0	0.287		0.2	0.385		0.4	0.812 -02		1.0	0.134
	7.0	0.739					10.0	0.280		0.3	0.222		0.6	0.510		2.0	0.641 -03
	10.0	0.729	10	0.56	0.964 -01		40.0	0.275		0.4	0.155		0.8	0.371		4.0	0.305
	30.0	0.720		0.7	0.720		50.0	0.275		0.6	0.958 -02		1.0	0.291		6.0	0.195
	50.0	0.720		1.0	0.463					0.8	0.691		2.0	0.138		8.0	0.142
				1.2	0.371	25	0.2	0.877 -01		1.0	0.539		3.0	0.886 -03		10.0	0.124
5	1.4	0.929 -01		2.0	0.201		0.3	0.484		1.2	0.441		4.0	0.645		14.0	0.114
	1.6	0.802		2.4	0.165		0.5	0.250		2.0	0.252		6.0	0.409		25.0	0.110
	2.0	0.658		3.0	0.139		0.8	0.143		4.0	0.117		7.0	0.364		50.0	0.110
	2.4	0.586		4.0	0.124		1.0	0.111		5.0	0.912 -03		8.0	0.341			
	3.0	0.532		5.0	0.118		1.2	0.902 -02		6.0	0.804		10.0	0.322	125	0.05	0.480 -01
	4.0	0.494		7.0	0.114		2.0	0.508		7.0	0.756		14.0	0.310		0.06	0.344
	5.0	0.478		10.0	0.112		3.0	0.323		8.0	0.731		40.0	0.306		0.08	0.210
	7.0	0.465		16.0	0.111		3.4	0.278		10.0	0.708		50.0	0.306		0.10	0.148
	10.0	0.459		50.0	0.111		4.0	0.235		16.0	0.692					0.14	0.917 -02
	30.0	0.454	15	0.34	0.968 -01		4.4	0.219		40.0	0.688	80	0.054	0.999 -01		0.2	0.578
	50.0	0.453		0.4	0.770		5.0	0.204		50.0	0.688		0.07	0.608		0.4	0.257
				0.6	0.191		6.0	0.191					0.09	0.391		0.6	0.165
6	1.2	0.837 -01		0.6	0.453		7.0	0.186	50	0.088	0.930 -01		0.10	0.328		0.8	0.121
	1.6	0.584		1.0	0.244		10.0	0.180		0.1	0.782		0.14	0.196		1.0	0.955 -03
	2.0	0.469		1.2	0.197		30.0	0.176		0.2	0.263		0.20	0.120		2.0	0.459
	2.4	0.411		2.0	0.109		50.0	0.176		0.3	0.154		0.24	0.950 -02		4.0	0.220
	3.0	0.369		2.4	0.090 -02					0.4	0.108		0.4	0.516		6.0	0.141
	4.0	0.341		3.0	0.601	30	0.16	0.904 -01		0.6	0.677 -02		0.6	0.328		9.0	0.904 -04
	5.0	0.329</															

PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

Interpolating in Table G.

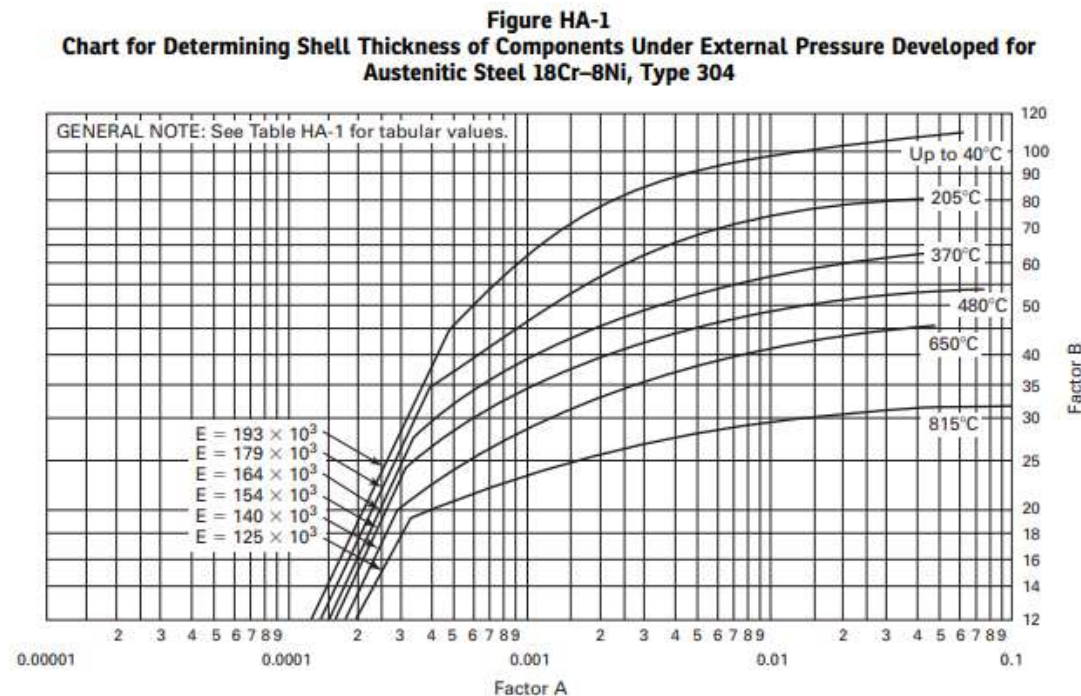
$$Do/t = 1016 \text{ mm} / 8.44 \text{ mm} = \underline{120.37 \text{ mm}},$$

$$L/Do = 18,000 \text{ mm} / 1016 \text{ mm} = \underline{17.72 \text{ mm}}$$

<i>Interpolation: ASME Section II, Part II Part D, Subpart 3, Figure G</i>								
Do/t	L/Do	A	Do/t	L/Do	A		Do/t	A
100	14	1.140E-04	125	14	7.400E-05		100	1.126E-04
	17.72	1.1265E-04		17.72	7.233E-05	A	120.37	7.980E-05
	25	1.100E-04		20	7.130E-05		125	7.233E-05

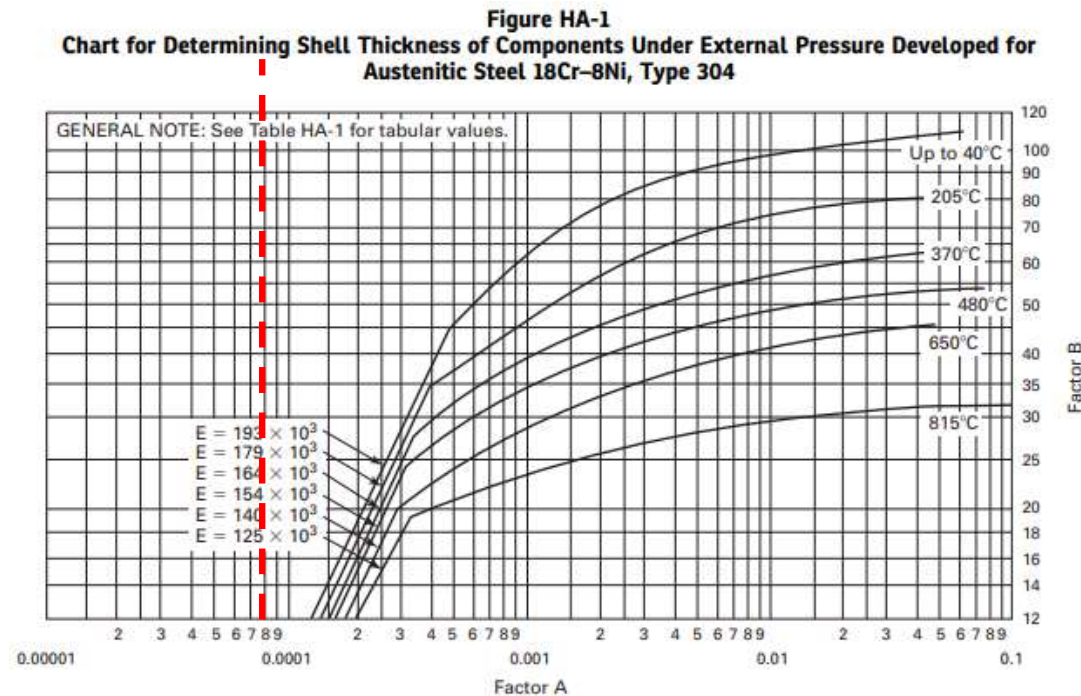
PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- Based on the figure or table, we obtain the value of factor A of 0.0000798. From the allowable stress table, we get the material figure corresponding to A358-304 material. Use Figure HA-1.



PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- Obtain the Factor B by plotting Factor A = 0.0000798.
- The value falls to the left of material-temperature line.



PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- **If the value of Factor B can be determined using the chart, use the following equation.**

$$Pa = \frac{4B}{3 \left(\frac{Do}{t} \right)}$$

- **If the factor A falls to the left of material –temperature chart, or the value of B cannot be determined, use the following equation.**

$$Pa = \frac{2AE}{3 \left(\frac{Do}{t} \right)}$$

PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- Calculate the Maximum Allowable External Design Pressure P_a .

$$P_a = \frac{2 \times 0.0000798 \times 1.95 \times 10^5 \text{ Mpa}}{3 \left(\frac{1016 \text{ mm}}{8.44 \text{ mm}} \right)} = 0.086 \text{ Mpa}$$

- $P_a < 0.10 \text{ Mpa} \rightarrow \text{“No Good”}$
- The result of the calculated maximum allowable external design pressure P_a is lower than the design external pressure P . Thus, the selected nominal thickness of the NPS 40 line is not enough to protect the system from a vacuum pressure of 0.10 Mpa.
- Select a pipe with a higher nominal thickness.

PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- **Re-Calculate using the nominal thickness of 12.7 mm**

$$t = 12.7 \text{ mm} - \text{MT} - \text{CA}$$

$$t = 12.7 \text{ mm} - 0.30 - 0$$

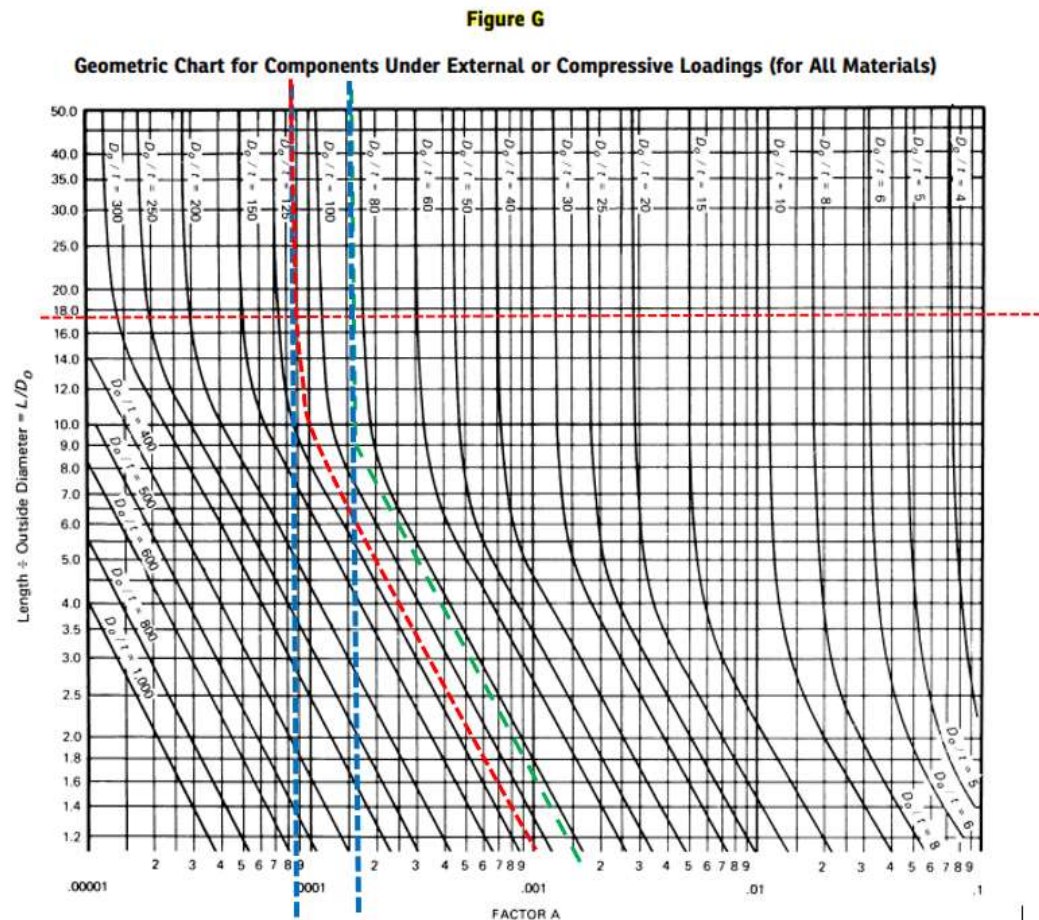
$$t = 12.4 \text{ mm}$$

$$D_o/t = 1016 \text{ mm} / 12.4 \text{ mm} = 81.935$$

$$L/D_o = 17.72 \text{ (same as previous calculation)}$$

PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- Obtaining the value of A from Figure G or Table G.



PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- Obtaining the value of A from Figure G or Table G.

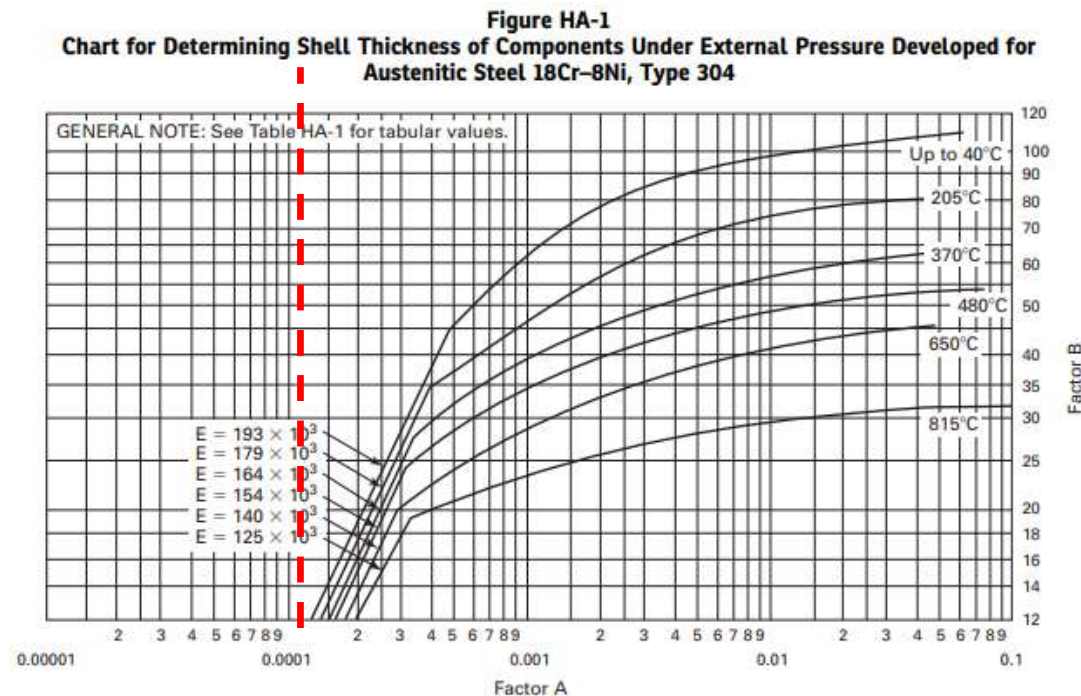
$$Do/t = 1016 \text{ mm} / 12.4 \text{ mm} = 81.935$$

$$L/Do = 17.72 \text{ (same as previous calculation)}$$

Interpolation: ASME Section II, Part II Part D, Subpart 3, Figure G								
Do/t	L/Do	A	Do/t	L/Do	A		Do/t	A
80	14	1.760E-04	100	14	1.140E-04	A	80	1.751E-04
	17.72	1.7507E-04		17.72	1.126E-04		81.9355	1.690E-04
	30	1.720E-04		25	1.100E-04		100	1.126E-04

PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- Obtain the Factor B by plotting Factor A = 0.00017.
- The value still falls to the left of material-temperature line.



PIPING WALL THICKNESS CALCULATION BASED ON EXTERNAL DESIGN PRESSURE

- The Maximum Allowable External Design Pressure P_a ,

$$P_a = \frac{2AE}{3 \left(\frac{D_o}{t} \right)} \quad P_a = \frac{2 \times 0.00017 \times 1.95E^5}{3 \left(\frac{1016}{12.40} \right)} = 0.269 \text{ Mpa}$$

- $P_a > 0.10 \text{ Mpa} \rightarrow \text{“Ok”}$
- The above result, can protect the piping system from a vacuum pressure of 0.10 Mpa. Select a thickness of 12.70 mm for the NPS 40 piping.

PIPING WALL THICKNESS CALCULATION RESULT:

NPS	OD [mm]	Pipe Material	Pressure Design Thickness, t [mm]	Minimum Required Thickness $t_m = t + CA$ [mm]	Nominal Thickness, $T = t_m + MT$ [mm]	Selected Nominal thickness [mm]	SCH No.	Validity check, $t < D/6$
40	1016	A358-304/304L	6.78	6.78	7.08	12.7		169.33
28	711	A358-304/304L	4.74	4.74	5.04	6.35	SCH 10S	118.50
24	610	A358-304/304L	4.07	4.07	4.37	6.35	SCH 10S	101.67
22	559	A358-304/304L	3.73	3.73	4.03	5.54	SCH 10S	93.17
20	508	A358-304/304L	3.39	3.39	3.69	5.54	SCH 10S	84.67
18	457	A358-304/304L	3.05	3.05	3.35	4.78	SCH 10S	76.17
16	406	A358-304/304L	2.71	2.71	3.01	4.76	SCH 10S	67.67
14	356	A358-304/304L	2.38	2.38	2.68	4.78	SCH 10S	59.33
12	324	A358-304/304L	2.16	2.16	2.46	4.57	SCH 10S	54.00
10	273	A358-304/304L	1.82	1.82	2.12	4.19	SCH 10S	45.50
8	219	A358-304/304L	1.46	1.46	1.76	3.76	SCH 10S	36.50

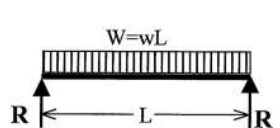
PIPE SUPPORT SPAN CALCULATION

- The piping system shall be designed with a proper support span in order to prevent undesirable sagging or deflection between supports.
- A properly supported piping will have a lower stress value that is very advantageous to the piping during operation. This span will be the basis of support placement in the design and construction stage of the project.

PIPE SUPPORT SPAN CALCULATION

TABLE 2.1
BASIC BEAM FORMULAS

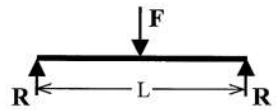
(a) Simple Beam – Uniform Load



$$R = \frac{wL}{2}, \quad M_{\max} = \frac{wL^2}{8}, \quad (\text{at Center})$$

$$\Delta_{\max} = \frac{5wL^4}{384EI}, \quad (\text{at Center})$$

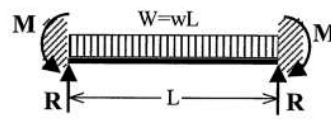
(b) Simple Beam – Concentrated Load



$$R = \frac{F}{2}, \quad M_{\max} = \frac{FL}{4}, \quad (\text{at Center})$$

$$\Delta_{\max} = \frac{FL^3}{48EI}, \quad (\text{at Center})$$

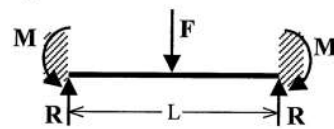
(c) Fixed Beam – Uniform Load



$$R = \frac{wL}{2}, \quad M_{\max} = \frac{wL^2}{12}, \quad (\text{at Ends})$$

$$\Delta_{\max} = \frac{wL^4}{384EI}, \quad (\text{at Center})$$

(d) Fixed Beam – Concentrated Load



$$R = \frac{F}{2}, \quad M_{\max} = \frac{FL}{8}, \quad (\text{at Ends and Center})$$

$$\Delta_{\max} = \frac{FL^3}{192EI}, \quad (\text{at Center})$$

MAX MOMENT EQUATION:

- $M_{\max} = wL^2/8 \rightarrow$ from (a)
- $M_{\max} = wL^2/12 \rightarrow$ from (c)

Take the average:

- $M_{\max} = wL^2/10$

MAX DEFLECTION EQUATION:

- $\Delta_{\max} = 5wL^4/384EI \rightarrow$ from (a)
- $\Delta_{\max} = wL^4/384EI \rightarrow$ from (c)

Take the average:

- $\Delta_{\max} = 2.5wL^4/384EI$

PIPE SUPPORT SPAN CALCULATION

Calculating the Pipe Span for 24" line.

- Allowable span based on stress, bending stress = ½ of hot allowable;

$$L1 = \sqrt{\frac{10ZS}{W}}$$

Where: Z = pipe section modulus
W = total linear weight
S = allowed bending stress

- Allowable span based on deflection, allowed deflection $\Delta = 12$ mm;

$$L2 = \sqrt[4]{\frac{128EI\Delta}{W}}$$

Where: E = Elastic Modulus
I = Moment of Inertia
 Δ = Deflection

PIPE SUPPORT SPAN CALCULATION

- **Solving for the Section Modulus and Moment of Inertia**

NPS 24

OD = 610 mm

Pipe Schedule = 10S

Thickness = 6.35 mm

Inside Diameter, ID = 610 – (2 x 6.35 mm) = 597.30 mm

Pipe Cross-section area = $\pi/4 (0.610^2 - 0.597.3^2) = 0.0122 \text{ m}^2$

Pipe Inside area, Ai = $\pi/4 \times (0.597)^2 = 0.28 \text{ m}^2$

$$Z = \frac{\pi(OD^4-ID^4)}{32 OD} = \frac{\pi(610^4-597.3^4)}{32 \times 610} = \mathbf{1,798,612 \text{ mm}^3}$$

$$I = \frac{\pi(OD^4-ID^4)}{64} = \frac{\pi(610^4-597.3^4)}{64} = \mathbf{548,576,604 \text{ mm}^4}$$

PIPE SUPPORT SPAN CALCULATION

- **Solving for the total linear weight, W.**

- Pipe Weight = $\rho_{\text{pipe}} \times A_{\text{pipe}} = 8030 \text{ kg/m}^3 \times 0.0122 \text{ m}^2 = 97.966 \text{ kg/m}$
- Fluid Weight = $\rho_{\text{water}} \times A_{\text{i,pipe}} = 1000 \text{ kg/m}^3 \times 0.28 \text{ m}^2 = 280 \text{ kg/m}$
- Insulation Weight = $\rho_{\text{insulation}} \times A_{\text{insulation}}$
 - Insulation thickness = 120 mm
 - Outside Diameter = $\text{OD}_{\text{pipe}} + 2 (\text{Insulation Thickness}) = 610 + (2 \times 120 \text{ mm}) = 850 \text{ mm}$
 - Cross-sectional Area of Insulation = $\pi/4 (0.850^2 - 0.610^2) = 0.275 \text{ m}^2$
- Insulation Weight = $32.8 \text{ kg/m}^3 \times 0.275 \text{ m}^2 = 9.03 \text{ kg/m}$
- Total Linear Weight, W = Pipe Weight + Fluid Weight + Insulation Weight
- $W = 97.966 + 280 + 9.03 = 387 \text{ kg/m} \Rightarrow \mathbf{3.796 \text{ N/mm}}$

PIPE SUPPORT SPAN CALCULATION

- **Solving for the Pipe Span Based on Stress, L1**

- Bending stress = ½ of Hot Allowable Stress = ½ x 137.9 Mpa
- Bending stress = 68.95 Mpa

$$L1 = \sqrt{\frac{10 \times 1,798,612 \times 68.95 \text{ Mpa}}{3.796 \text{ N/mm}}} = 18,074 \text{ mm} = 18.074 \text{ m}$$

- **Solving for the Pipe Span Based on Deflection, L2**

- Allowed Deflection is 12 mm

$$L2 = \sqrt[4]{\frac{128 \times 195 \text{E}^5 \text{Mpa} \times 548,576,603 \text{ mm}^4 \times 12 \text{ mm}}{3.796 \text{ N/mm}}} = 14,424 \text{ mm} = \mathbf{14.4 \text{ m}}$$

PIPE SUPPORT SPAN CALCULATION

Calculating the Pipe Span for 10" line.

NPS 10

OD = 273 mm

Pipe Schedule = 10S

Thickness = 4.19 mm

Inside Diameter, ID = $273 - (2 \times 4.19 \text{ mm}) = 264.62 \text{ mm}$

Pipe Cross-section area = $\pi/4 (0.273^2 - 0.2646^2) = 0.0035 \text{ m}^2$

Pipe Inside area, $A_i = \pi/4 \times (0.26462)^2 = 0.055 \text{ m}^2$

Section Modulus, $Z = \mathbf{234,198 \text{ mm}^3}$

Moment of Inertia $I = \mathbf{31,968,015 \text{ mm}^4}$

PIPE SUPPORT SPAN CALCULATION

- **Solving for the Total Linear Weight, W.**

- Pipe Weight = $\rho_{\text{pipe}} \times A_{\text{pipe}} = 8030 \text{ kg/m}^3 \times 0.00354 \text{ m}^2 = 28.4 \text{ kg/m}$
- Fluid Weight = $\rho_{\text{water}} \times A_{\text{i,pipe}} = 1000 \text{ kg/m}^3 \times 0.055 \text{ m}^2 = 55 \text{ kg/m}$
- Insulation Weight = $\rho_{\text{insulation}} \times A_{\text{insulation}}$
 - Insulation thickness = 100 mm
 - Outside Diameter = $\text{OD}_{\text{pipe}} + 2 (\text{Insulation Thickness}) = 273 + (2 \times 100 \text{ mm}) = 473 \text{ mm}$
 - Cross-sectional Area of Insulation = $\pi/4 (0.473^2 - 0.273^2) = 0.117 \text{ m}^2$
- Insulation Weight = $32.8 \text{ kg/m}^3 \times 0.117 \text{ m}^2 = 3.84 \text{ kg/m}$
- Total Linear Weight, W = Pipe Weight + Fluid Weight + Insulation Weight
- $W = 28.4 + 55 + 3.84 = 87.24 \text{ kg/m} \Rightarrow \mathbf{0.86 \text{ N/mm}}$

PIPE SUPPORT SPAN CALCULATION

- **Solving for the Pipe Span Based on Stress, L1**

- Bending stress = $\frac{1}{2}$ of Hot Allowable Stress = $\frac{1}{2} \times 137.9 \text{ Mpa}$
- Bending stress = 68.95 Mpa

$$L1 = \sqrt{\frac{10 \times 234,198 \times 68.95 \text{ Mpa}}{0.86 \text{ N/mm}}} = 13,702.8 \text{ mm} = 13.702 \text{ m}$$

- **Solving for the Pipe Span Based on Deflection, L2**

- Allowed Deflection is 12 mm

$$L2 = \sqrt[4]{\frac{128 \times 195 \text{E}^5 \text{Mpa} \times 31,968,015 \text{ mm}^4 \times 12 \text{ mm}}{0.86 \text{ N/mm}}} = 10,272.14 \text{ mm} = \mathbf{10.3 \text{ m}}$$

PIPE SUPPORT SPAN CALCULATION

SIZE	SCH	INSULATION THICKNESS	INSULATION DENSITY	L1	L2	ALLOWED MAX PIPE SPAN
4"	10S	80 mm	32.8 kg/m ³	10.23	7.14	7 m
6"	10S	100 mm	32.8 kg/m ³	11.56	8.36	8 m
8"	10S	100 mm	32.8 kg/m ³	12.67	9.35	9 m
10"	10S	100 mm	32.8 kg/m ³	13.73	10.29	10 m
12"	10S	110 mm	32.8 kg/m ³	14.56	11.06	11 m
14"	10S	110 mm	32.8 kg/m ³	16.54	12.06	12 m
16"	10S	120 mm	32.8 kg/m ³	15.31	12.00	12 m
18"	10S	120 mm	32.8 kg/m ³	15.61	12.48	12 m
20"	10S	120 mm	32.8 kg/m ³	16.75	13.27	13 m
24"	10S	130 mm	32.8 kg/m ³	18.07	14.43	14 m
28"	10S	130 mm	32.8 kg/m ³	20.05	15.79	15 m
40"	12.7	130 mm	32.8 kg/m ³	21.80	18.00	18 m

PIPING HYDRAULIC CALCULATION

- Mass Flow Rate = 94.5 kg/sec
- Total Flow Requirement = 740 m³/hour
- Fluid Density = 461.6 kg/m³
- Friction Factor Contingency = 15%
- Buffer Tank Inlet Pressure = 0.10 Mpa
- Buffer Tank Nozzle Elevation = 9 m
- Absolute Roughness = 0.045 mm
- Dynamic Viscosity = 0.147 cP = 0.000147 N-s/m²

PIPING HYDRAULIC CALCULATION

Marine Loading Arm (MLA)



PIPING HYDRAULIC CALCULATION

Section A: Cargo Pump to Marine Loading Arm.

Pipe Size = 16"

Pipe Schedule = 10S

Pipe Internal Diameter = 0.396 m

Pipe Internal Area = 0.124 m²

Total Pipe Developed Length = 35+25+15+15 = 90 m

Total 90°LR Elbow = 13 Elbows

Elevation Difference = 17.6 – (-11 m) = 28.6 m

PIPING HYDRAULIC CALCULATION

Section A: Cargo Pump to Marine Loading Arm.

- Equivalent Length:

Item	Quantity	Unit Le, m	Le, m
Straight Pipe	90	1	90
Elbows, 90°, LR	13	6.1	79.3
Elbows, 45°, LR			
Tee			
Reducer			
TOTAL			169.3

Consider a 25% margin to account for some future adjustments.

Total Length = 169.3 m x 1.25 = **211.625 m**

PIPING HYDRAULIC CALCULATION

Section A: Cargo Pump to Marine Loading Arm.

- Solving for the Fluid Velocity:

$$V_a = Q/A$$

$$V_a = m/\rho A$$

$$V_a = 94.9 \text{ kg/s} / (461.6 \text{ kg/m}^3 \times 0.124 \text{ m}^2)$$

$$V_a = 1.66 \text{ m/s} \rightarrow \text{velocity check "ok"}$$

PIPING HYDRAULIC CALCULATION

Section A: Cargo Pump to Marine Loading Arm.

- Solving for the Reynolds Number:

$$Re_a = \rho V D h / \mu$$

$$Re_a = [461.6 \text{ kg/m}^3 \times 1.66 \text{ m/s} \times 0.396 \text{ m}] / 1.47\text{E-}4 \text{ N-s/m}^2$$

$$Re_a = 2,064,199.8$$

$$Re_a = \text{Turbulent flow}$$

PIPING HYDRAULIC CALCULATION

Section A: Cargo Pump to Marine Loading Arm.

- Solving for the Relative Roughness:

Relative roughness = ε / D_h

Absolute roughness for Commercial Steel Pipe = 0.045 mm
(welded)

Relative Roughness = 0.045 mm / 396 mm

Relative Roughness = **0.0001136**

PIPING HYDRAULIC CALCULATION

Section A: Cargo Pump to Marine Loading Arm.

- Obtaining Friction Factor using Moody Chart: $f = 0.013$

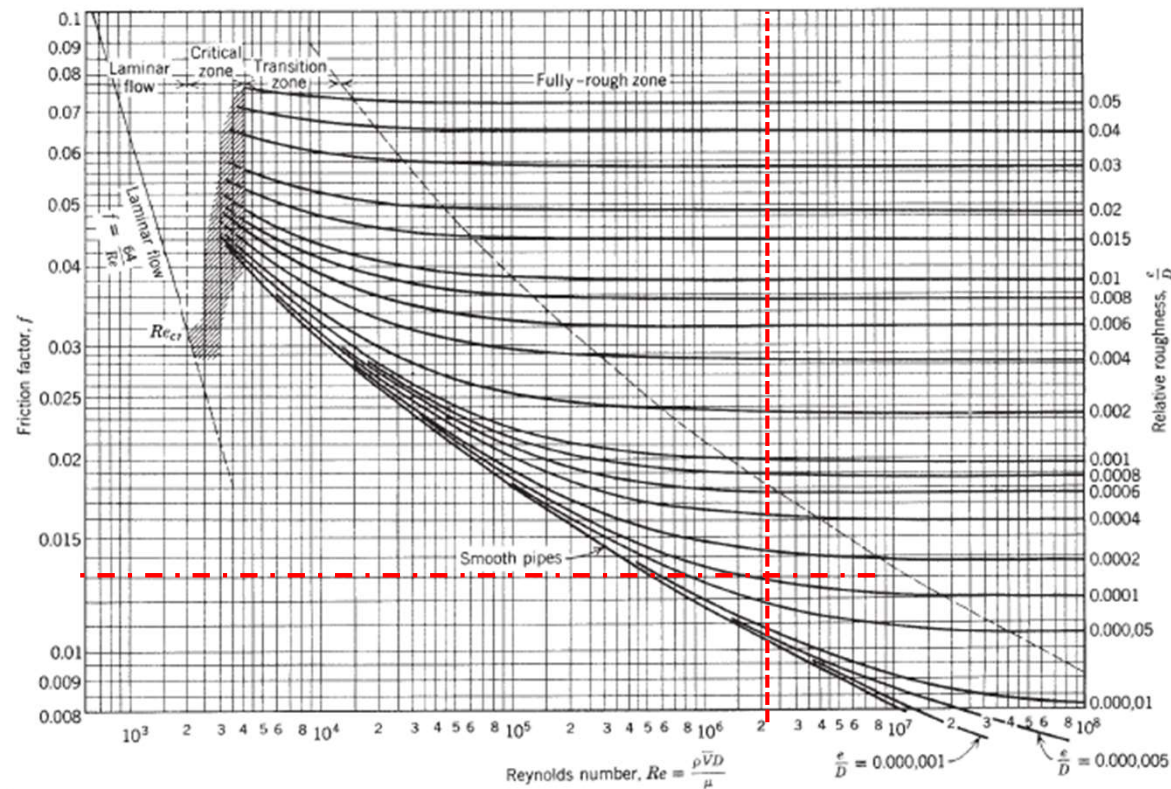


FIGURE 11.1 Moody Diagram, L.F. Moody, "Friction factors for pipe flow," Trans. ASME Vol. 66, 1944. Reprinted by permission of The American Society of Mechanical Engineers. All rights reserved.

PIPING HYDRAULIC CALCULATION

Section A: Cargo Pump to Marine Loading Arm.

- Solving Head Loss due to Friction

$$Hf_a = f \frac{L_e V^2}{2 D_h g_c}$$

$$Hf_a = 0.013 \times 211.625 \text{ m} \times (1.66 \text{ m/s})^2 / 2 \times 0.396 \text{ m} \times 9.81 \text{ m/s}^2$$

$$Hf_a = \mathbf{0.973 \text{ m}}$$

In terms of pressure;

$$Hf_a = 0.973 \text{ m} \times 9.81 \text{ m/s}^2 \times 461.6 \text{ kg/m}^3 / 1000$$

$$Hf_a = 4.406 \text{ Kpa} = \mathbf{0.0044 \text{ Mpa}}$$

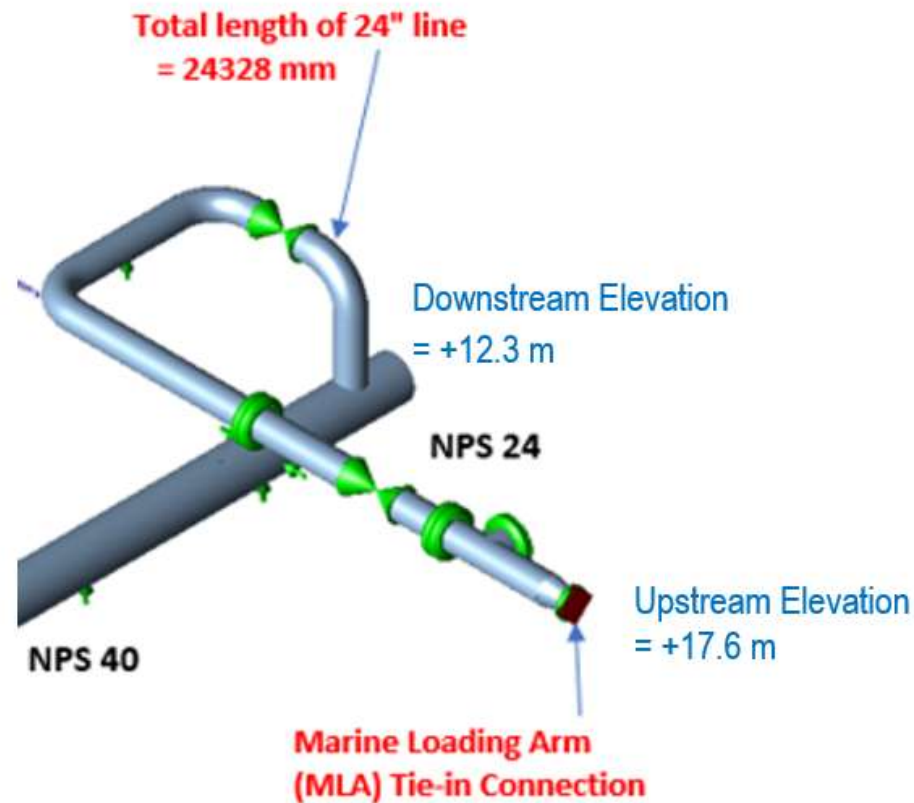
PIPING HYDRAULIC CALCULATION

Section A: Cargo Pump to Marine Loading Arm.

- Solving Elevation Head, Z_a
 - Upstream Elevation = -11 m (below cD)
 - Downstream Elevation = +17.6 m (above cD)
 - $Z_a = 17.6 \text{ m} - (-11 \text{ m}) = 28.6 \text{ m}$
- Elevation Head, in terms of Pressure
 - $Z_a = \rho gh = (461.6 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times 28.6 \text{ m}) / 1000$
 - $Z_a = 129.51 \text{ Kpa}$
 - $Z_a = 0.130 \text{ Mpa}$

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)



PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

Pipe Size = 24"

Pipe Schedule = 10S

Pipe Internal Diameter = 0.5969 m

Pipe Internal Area = 0.28 m²

Total Pipe Developed Length = 24.328 m

Total 90°LR Elbow = 3 Elbows

Elevation Difference = 17.6 – 12.3 m = 5.30 m

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

- Equivalent Length:

Item	Quantity	Unit Le, m	Le, m
Straight Pipe	24.328	1	24.328
Elbows, 90°, LR	3	9.2	27.6
Elbows, 45°, LR			
Tee	1	34.4	34.4
Reducer			
TOTAL			86.328

Consider a 25% margin to account for unforeseen route adjustment.

Total Length = 86.328 m x 1.25 = **107.91 m**

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

- Solving for the Fluid Velocity:

$$V_b = Q/A$$

$$V_b = m/\rho A$$

$$V_b = 94.9 \text{ kg/s} / (461.6 \text{ kg/m}^3 \times 0.28 \text{ m}^2)$$

$$V_b = \mathbf{0.73 \text{ m/s}} \rightarrow \text{velocity check "ok"}$$

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

- Solving for the Reynolds Number:

$$Re_b = \rho V D h / \mu$$

$$Re_b = [461.6 \text{ kg/m}^3 \times 0.73 \text{ m/s} \times 0.5969 \text{ m}] / 1.47\text{E}^{-4} \text{ N-s/m}^2$$

$$Re_b = 1,368,273.5$$

$$Re_b = \text{Turbulent flow}$$

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

- Solving for the Relative Roughness:

Relative roughness = ε / D_h

Absolute roughness for Commercial Steel Pipe = 0.045 mm
(welded)

Relative Roughness = 0.045 mm / 596.9 mm

Relative Roughness = **0.000075**

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

Obtaining Friction Factor using Moody Chart: $f = 0.0127$

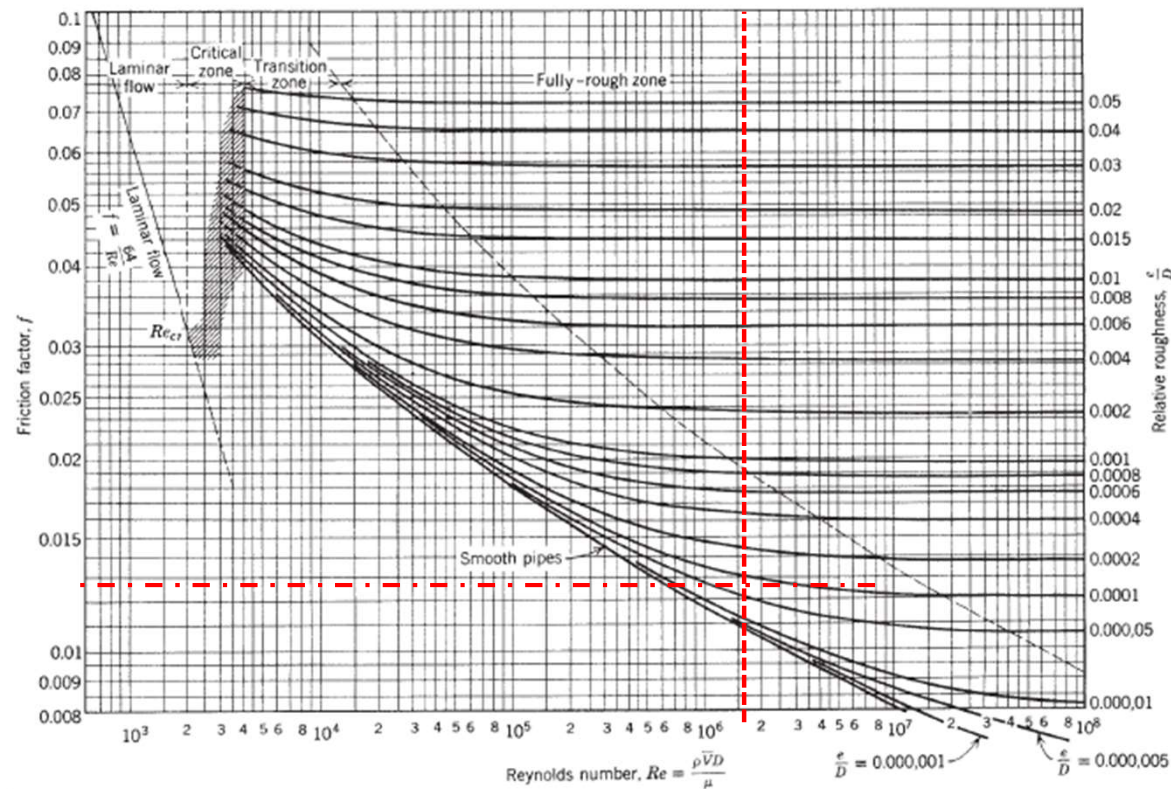


FIGURE 11.1 Moody Diagram, L.F. Moody, "Friction factors for pipe flow," Trans. ASME Vol. 66, 1944. Reprinted by permission of The American Society of Mechanical Engineers. All rights reserved.

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

- Solving Head Loss due to Friction

$$H_{fb} = f \frac{L_e V^2}{2 D_h g_c}$$

$$H_{fb} = 0.0127 \times 107.91 \text{ m} \times (0.73 \text{ m/s})^2 / 2 \times 0.5969 \text{ m} \times 9.81 \text{ m/s}^2$$

$$H_{fb} = \mathbf{0.063 \text{ m}}$$

In terms of pressure;

$$H_{fb} = 0.063 \text{ m} \times 9.81 \text{ m/s}^2 \times 461.6 \text{ kg/m}^3 / 1000$$

$$H_{fb} = 0.285 \text{ Kpa} = \mathbf{0.000285 \text{ Mpa}}$$

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

- Solving Elevation Head, Z_a
 - Upstream Elevation = +12.3 m (above cD)
 - Downstream Elevation = +17.6 m (above cD)
 - $Z_b = 12.3 \text{ m} - 17.6 \text{ m} = -5.3 \text{ m}$
- Elevation Head, in terms of Pressure
 - $Z_b = \rho gh = (461.6 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times -5.3 \text{ m}) / 1000$
 - $Z_b = -24 \text{ Kpa}$
 - $Z_b = -0.024 \text{ Mpa}$

PIPING HYDRAULIC CALCULATION

Section B: MLA Tie-in to 40" Header (24" Line)

- Additional Pressure Drop of Piping Elements.

Item	Qty	Pressure Drop, Mpa	Total Pressure Drop, Mpa
Check Valve	1	0.005	0.005
Filter / Strainer	1	0.005	0.005
Isolation Valve	2	0.005	0.01
Total			0.02

PIPING HYDRAULIC CALCULATION

Section C: 40" Header

Pipe Size = 40"

Pipe Thickness = 12.7 mm

Pipe Internal Diameter = 0.9906 m

Pipe Internal Area = 0.771 m²

Total Pipe Developed Length = 16.495 m

Total 90°LR Elbow = 0 Elbows

Tee = 1 pc

Elevation Difference = 12.3 – 12.3 m = 0 m

PIPING HYDRAULIC CALCULATION

Section C: 40" Header

- Equivalent Length:

Item	Quantity	Unit Le, m	Le, m
Straight Pipe	16.495	1	16.495
Elbows, 90°, LR			
Elbows, 45°, LR			
Tee	1	58.7	58.7
Reducer			
TOTAL			75.195

Consider a 25% margin to account for unforeseen route adjustment.

Total Length = 75.195 m x 1.25 = **93.99 m**

PIPING HYDRAULIC CALCULATION

Section C: 40" Header

- Solving for the Fluid Velocity:

$$V_c = Q/A$$

$$V_c = m/\rho A$$

$$V_c = 94.9 \text{ kg/s} / (461.6 \text{ kg/m}^3 \times 0.771 \text{ m}^2)$$

$$V_c = \mathbf{0.27 \text{ m/s}} \rightarrow \text{velocity check "ok"}$$

PIPING HYDRAULIC CALCULATION

Section C: 40" Header

- Solving for the Reynolds Number:

$$Re_c = \rho V D h / \mu$$

$$Re_c = [461.6 \text{ kg/m}^3 \times 0.27 \text{ m/s} \times 0.9906 \text{ m}] / 1.47\text{E-}^4 \text{ N-s/m}^2$$

$$Re_c = 829,455$$

$$Re_c = \text{Turbulent flow}$$

PIPING HYDRAULIC CALCULATION

Section C: 40" Header

- Solving for the Relative Roughness:

Relative roughness = ϵ / D_h

Absolute roughness for Commercial Steel Pipe = 0.045 mm
(welded)

Relative Roughness = 0.045 mm / 990.6 mm

Relative Roughness = **0.000045**

PIPING HYDRAULIC CALCULATION

Section C: 40" Header

Obtaining Friction Factor using Moody Chart: $f = 0.0129$

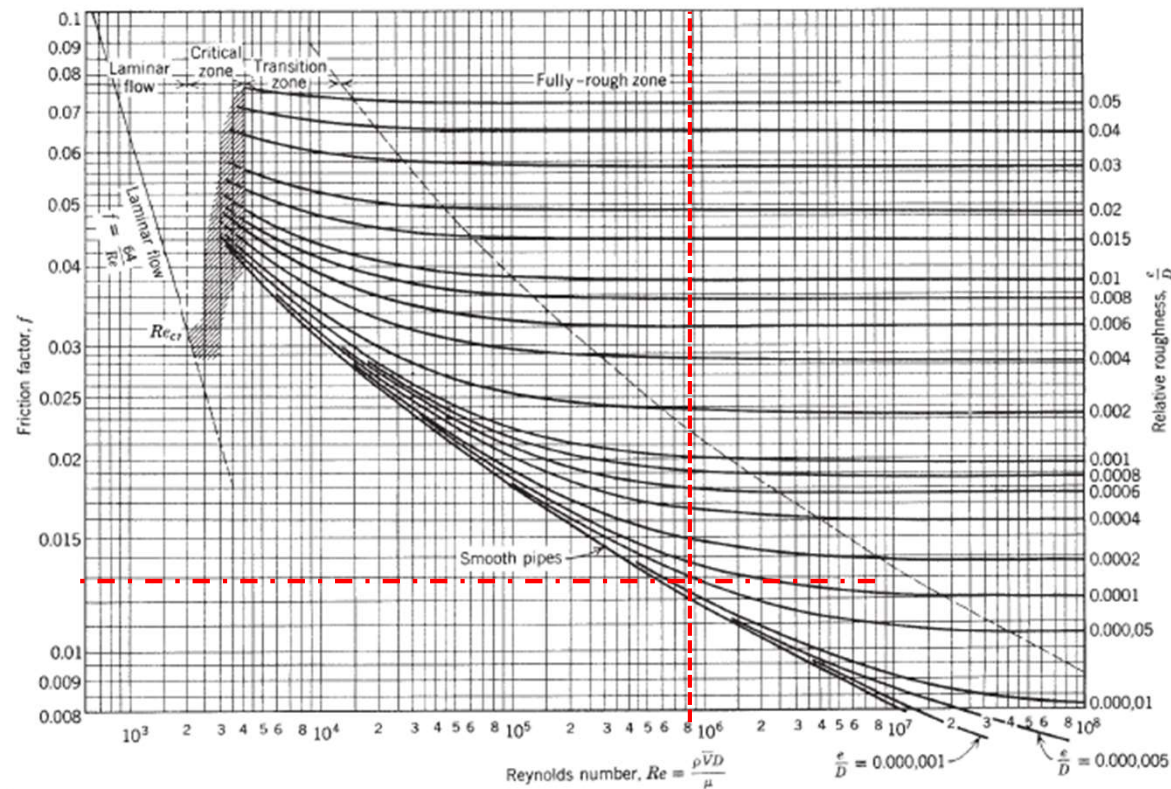


FIGURE 11.1 Moody Diagram, L.F. Moody, "Friction factors for pipe flow," Trans. ASME Vol. 66, 1944. Reprinted by permission of The American Society of Mechanical Engineers. All rights reserved.

PIPING HYDRAULIC CALCULATION

Section C: 40" Header

- Solving Head Loss due to Friction

$$Hf_c = f \frac{L_e V^2}{2 D_h g_c}$$

$$Hf_c = 0.0129 \times 93.99 \text{ m} \times (0.27 \text{ m/s})^2 / 2 \times 0.9906 \text{ m} \times 9.81 \text{ m/s}^2$$

$$Hf_c = \mathbf{0.004 \text{ m}}$$

In terms of pressure;

$$Hf_c = 0.004 \text{ m} \times 9.81 \text{ m/s}^2 \times 461.6 \text{ kg/m}^3 / 1000$$

$$Hf_c = 0.020 \text{ Kpa} = \mathbf{0.000020 \text{ Mpa}}$$

PIPING HYDRAULIC CALCULATION

Section C: 40" Header

- Solving Elevation Head, Z_a
 - Upstream Elevation = +12.3 m (above cD)
 - Downstream Elevation = +12.3 m (above cD)
 - $Z_c = 12.3 \text{ m} - 12.3 \text{ m} = 0 \text{ m}$
- Elevation Head, in terms of Pressure
 - $Z_c = \rho gh = (461.6 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times 0 \text{ m}) / 1000$
 - $Z_c = 0 \text{ Kpa}$
 - $Z_c = 0 \text{ Mpa}$

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

Pipe Size = 10"

Pipe Schedule = 10S

Pipe Internal Diameter = 0.2647 m

Pipe Internal Area = 0.055 m²

Total Pipe Developed Length = 1581.318 m

Total 90°LR Elbow = 64 Elbows

Total 45°LR Elbow = 10 Elbows

Elevation Difference = 9 m – 12.3 = -3.3 m

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

- Equivalent Length:

Item	Quantity	Unit Le, m	Le, m
Straight Pipe	1581.318	1	1581.318
Elbows, 90°, LR	64	4.1	262.4
Elbows, 45°, LR	10	3.35	33.5
Tee			
Reducer			
TOTAL			1877.218

Consider a 25% margin to account for unforeseen route adjustment.

Total Length = 1877.218 m x 1.25 = **2346.5225 m**

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

- Solving for the Fluid Velocity:

$$V_d = Q/A$$

$$V_d = m/\rho A$$

$$V_d = 94.9 \text{ kg/s} / (461.6 \text{ kg/m}^3 \times 0.055 \text{ m}^2)$$

$$V_d = 3.74 \text{ m/s} \rightarrow \text{velocity check "ok"}$$

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

- Solving for the Reynolds Number:

$$Re_d = \rho V D h / \mu$$

$$Re_d = [461.6 \text{ kg/m}^3 \times 3.74 \text{ m/s} \times 0.2647 \text{ m}] / 1.47E^{-4} \text{ N-s/m}^2$$

$$Re_d = 3,106,992$$

$$Re_d = \text{Turbulent flow}$$

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

- Solving for the Relative Roughness:

Relative roughness = ε / D_h

Absolute roughness for Commercial Steel Pipe = 0.045 mm
(welded)

Relative Roughness = 0.045 mm / 264.7 mm

Relative Roughness = **0.00017**

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

Obtaining Friction Factor using Moody Chart: $f = 0.0137$

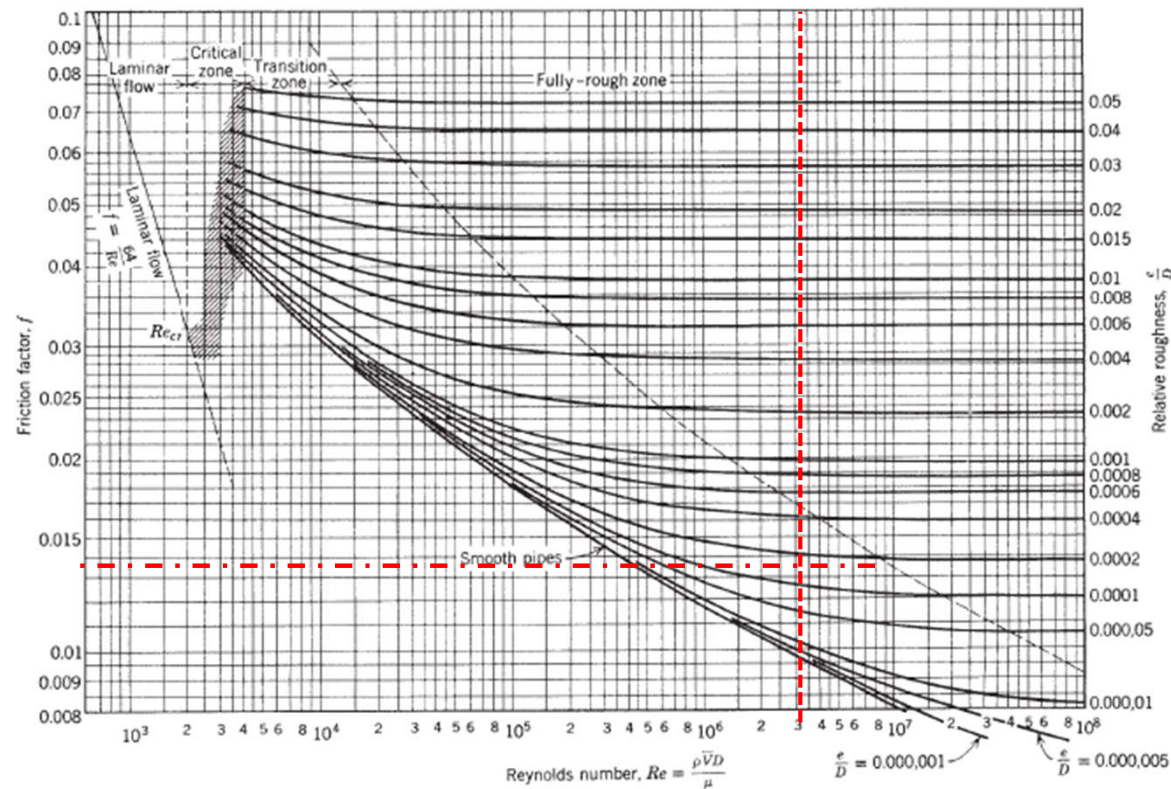


FIGURE 11.1 Moody Diagram, L.F. Moody, "Friction factors for pipe flow," Trans. ASME Vol. 66, 1944. Reprinted by permission of The American Society of Mechanical Engineers. All rights reserved.

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

- Solving Head Loss due to Friction

$$H_{fd} = f \frac{L_e V^2}{2 D_h g_c}$$

$$H_{fd} = 0.0137 \times 2346.5225 \text{ m} \times (3.74 \text{ m/s})^2 / 2 \times 0.2647 \text{ m} \times 9.81 \text{ m/s}^2$$

$$H_{fd} = \mathbf{86.49 \text{ m}}$$

In terms of pressure;

$$H_{fd} = 86.49 \text{ m} \times 9.81 \text{ m/s}^2 \times 461.6 \text{ kg/m}^3 / 1000$$

$$H_{fd} = 391.654 \text{ Kpa} = \mathbf{0.392 \text{ Mpa}}$$

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

- Solving Elevation Head, Z_d
 - Upstream Elevation = +12.3 m (above cD)
 - Downstream Elevation = +9.0 m (above cD)
 - $Z_c = 9.0 \text{ m} - 12.3 \text{ m} = -3.3 \text{ m}$
- Elevation Head, in terms of Pressure
 - $Z_c = \rho gh = (461.6 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times -3.3 \text{ m}) / 1000$
 - $Z_c = -14.94 \text{ Kpa}$
 - $Z_c = -0.015 \text{ Mpa}$

PIPING HYDRAULIC CALCULATION

Section D: 10" Unloading Line to Buffer Tank

Additional Pressure Drop of Piping Elements.

Item	Qty	Pressure Drop, Mpa	Total Pressure Drop, Mpa
Venturi meter	1	0.005	0.005
Filter / Strainer			
Isolation Valve	4	0.005	0.02
Total			0.025

PIPING HYDRAULIC CALCULATION

Total Head Loss due to friction, H_{ft}

- $H_{ft} = H_{fa} + H_{fb} + H_{fc} + H_{fd}$
- $H_{ft} = 0.973 + 0.063 + 0.004 + 86.49 = \mathbf{87.531\ m}$
- $H_{ft} = 87.531\ m \times 461.6\ kg/m^3 \times 9.81\ m/s^2$
- $H_{ft} = 396,366.3\ Pa = \mathbf{0.3963\ Mpa}$

Total Static Head Loss, Z_t

- $Z_t = Z_a + Z_b + Z_c + Z_d$
- $Z_t = 28.6 + (-5.3) + 0 + (-3.3) = \mathbf{20\ m}$
- $Z_t = 20\ m \times 461.6\ kg/m^3 \times 9.81\ m/s^2$
- $Z_t = 90565.95\ Pa = \mathbf{0.0905\ Mpa}$

PIPING HYDRAULIC CALCULATION

Total Pressure Drop of Piping Elements, P_{DE}

Pressure Drop of Pipe elements at Section B = 0.02 Mpa

Pressure Drop of Pipe elements at Section D = 0.025 Mpa

$$P_{DE} = 0.02 \text{ Mpa} + 0.025 \text{ Mpa} = 0.045 \text{ Mpa}$$

Total Pressure Drop of the Piping System, P_D

$$P_D = H_{ft} + Z_t + P_{DE} + \text{Buffer Tank Inlet Pressure}$$

$$P_D = 0.3963 \text{ Mpa} + 0.0905 \text{ Mpa} + 0.045 \text{ Mpa} + 0.10 \text{ Mpa}$$

$$P_D = 0.632 \text{ Mpa}$$

$$P_D = 139.567 \text{ m}$$

PIPING HYDRAULIC CALCULATION

- In practice, we always provide 15% factor of safety. This factor will take the possibility of pipe aging as well as contingency for unknown factors.

$$P_D = 0.632 \text{ Mpa} \times 1.15 = 0.727 \text{ Mpa or } 160.5 \text{ m}$$

Pump Head Adequacy Check

Unit	Pump Total Head	Calculated Pressure Drop	Remarks
Meters	190 m	160.502 m	OK
MPa	0.86 MPa	0.727 MPa	OK

PIPING FLEXIBILITY AND STRESS ANALYSIS

The following design parameter are used as an input value to the stress analysis software CAESAR II.

Size	Schedule	Material	CA,mm	MT	Insulation Thickness	Insulation Density
40"	12.7 mm	A358-304	0	EFW	130 mm	32.8 kg/m ³
24"	10S	A358-304	0	EFW	120 mm	32.8 kg/m ³
10"	10S	A358-304	0	EFW	100 mm	32.8 kg/m ³

PIPING FLEXIBILITY AND STRESS ANALYSIS

The following design parameter are used as an input value to the stress analysis software CAESAR II.

Information	Value
Design Pressure	18.5 barg (1.85 N/mm ²)
Operating Pressure	5.0 barg (0.50 N/mm ²)
Design Temperature	80°C / -172°C
Operating Temperature	-160°C
Fluid Density	461.6 kg/m ³
Pipe Density	8030 kg/m ³

Classic Piping Input

From: 30 To: 40 ☐ Name

DX: -1557.000 mm DY: DZ: ☐ Offsets

Diameter: 609.6000 WYSch: 6.3500 ☐ Seam Welded Wt Factor: 1.0000 -Mill Tol %: 4.7200 Corrosion: 0.0000 Pipe Den: 8027.19971 Fluid Den 1: 461.50000 Fluid Den 2: Hydro Den:

Temp 1: 80.0000 Temp 2: -172.0000 Temp 3: -160.0000 Pressure 1: 18.5000 Pressure 2: Pressure 3: Hydro Press: 27.7500

☐ Bend ☐ Rigid ☐ Expansion Joint ☐ Reducer ☐ SIFs & Tees ☐ Restraints ☐ Displacements ☐ Hangers ☐ Flange Checks ☐ Nozzle Flex. ☐ Nozzle Lmt Check ☐ Forces/Moments ☐ Uniform Loads ☐ Wind / Wave

Material: (192)A358 304 ☐ Allowable Stress

Elastic Modulus (C): 1.9459E+05 Elastic Modulus (H1): 1.9062E+05 Elastic Modulus (H2): 2.0734E+05 Elastic Modulus (H3): 2.0662E+05 Poisson's Ratio: 0.2920

Refract Thk: Refract Density: Insul Thk: 130.0000 Clad Thk: 50.0000 Insulation Density: 45.00000 Cladding Density: 120.00000 Insul/Cladding Unit Weight:

Model Status: Allowable Stresses Bends Displacements Equipment Limits

Current Element: 3 Total Elements: 164/32000

of Node Names: 0/13350 # of Bends: 21/17600 # of Rigid: 42/17600 # of Exp. Joints: 0/17600 # of Restraints: 28/17600 # of Displacements: 0/17600 # of Uniform Loads: 1/17600 # of Flanges: 0/13350 # of Eqp Limits: 0/13350 # of Forces/Moments: 10/17600 # of Wind Specs: 1/17600 # of Element Offsets: 0/17600 # of Allow Stress: 5/17600 # of Reducers: 4/13350 # of SIFs & Tees: 10/17600 # of Hangers: 0/17600 # of Nozzles: 0/13350

Line Number: 129101-LNG-24 Element Name:

PIPING FLEXIBILITY AND STRESS ANALYSIS

Pipe Span Check:

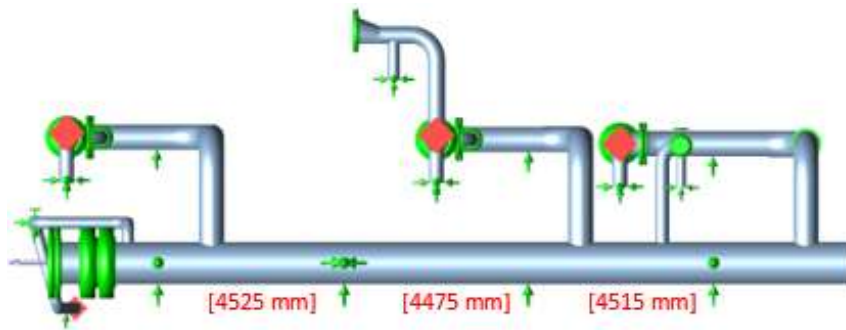


Figure. Pipe Span Check for 40" Line



- Allowed Span for 40" = 18 m
- Allowed Span for 10" = 10 m
- The span for both 40" and 10" line are within the maximum allowed span.

PIPING FLEXIBILITY AND STRESS ANALYSIS

SUSTAINED STRESS

Calculation of Sustained Stress:

The location of highest sustained stress is at node 450.

Pipe Diameter = 609.6 mm

$S_a = SLP + F/A$; $F/A \rightarrow$ can be neglected

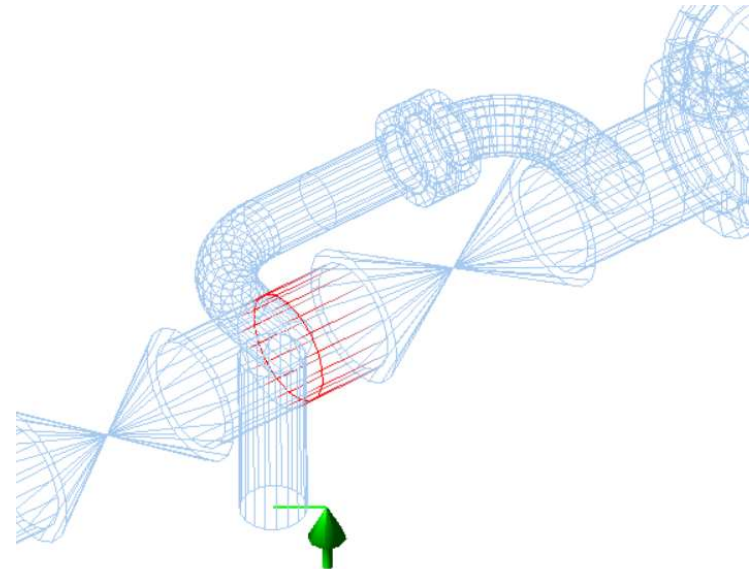
$SLP = PD/4t$

$SLP = [1.8\text{Mpa} \times 609.6\text{mm}] / [4 \times 6.35]$

SLP = 43.2 Mpa

Bending Stress, **Sb** = 148.7 Mpa (by CII)

Torsional Stress, **St** = -0.30 Mpa (by CII)



PIPING FLEXIBILITY AND STRESS ANALYSIS

SUSTAINED STRESS

Calculation of Sustained Stress:

$$S_L = \sqrt{[S_a + S_b]^2 + (2S_t)^2}$$

$$S_L = \sqrt{[43.2 + 148.7]^2 + (2 \times 0.30)^2}$$

$$S_L = 191.9 \text{ Mpa}$$

Evaluation:

$$S_L > S_H$$

$$191.9 \text{ Mpa} > 137.9 \text{ Mpa} \rightarrow \text{"Failed"}$$

Calculation of Allowable Stress:

For A358-304, from ASME B31.3

$$S_Y = 30 \text{ ksi} = 206.84 \text{ Mpa}$$

$$S_U = 75 \text{ ksi} = 517.10 \text{ Mpa}$$

$$S_H = S_Y / 1.5 = 206.84 \text{ Mpa} / 1.5$$

$$S_H = 137.9 \text{ Mpa}$$

$$S_H = S_U / 3.0 = 517.10 \text{ mpa} / 3.0$$

$$S_H = 172.35 \text{ Mpa}$$

Use the lower value,

$$S_H = 137.9 \text{ Mpa}$$

PIPING FLEXIBILITY AND STRESS ANALYSIS

OCCASIONAL STRESS

Calculation of Occasional Stress:

The location of highest sustained stress is at node 450.

Pipe Diameter = 609.6 mm

$S_a = SLP + F/A$; $F/A \rightarrow$ can be neglected

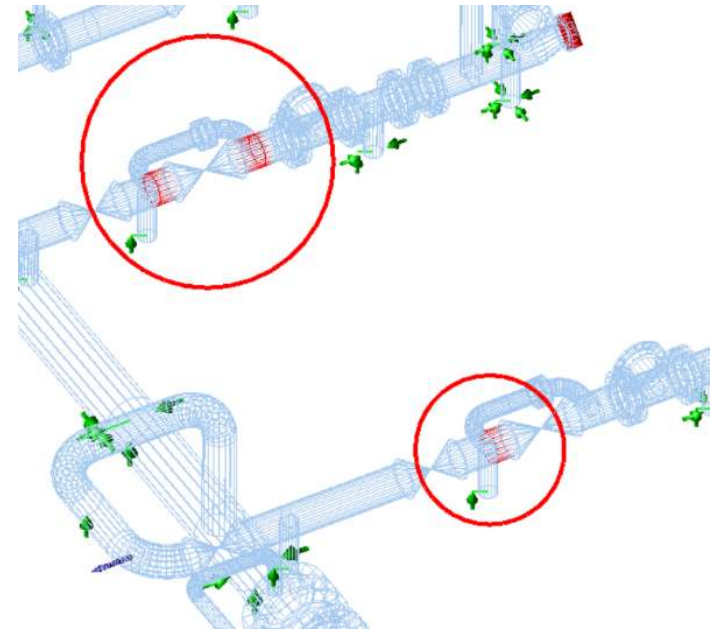
$SLP = PD/4t$

$SLP = [1.8\text{Mpa} \times 609.6\text{mm}] / [4 \times 6.35]$

SLP = 43.2 Mpa

Bending Stress, **S_b** = 266.9 Mpa (by CII)

Torsional Stress, **S_t** = 0.40 Mpa (by CII)



PIPING FLEXIBILITY AND STRESS ANALYSIS

OCCASIONAL STRESS

Calculation of Occasional Stress:

$$S_L = \sqrt{[S_a + S_b]^2 + (2S_t)^2}$$

$$S_L = \sqrt{[43.2 + 266.9]^2 + (2 \times 0.40)^2}$$

$$S_L = 310.40 \text{ Mpa}$$

Evaluation:

$$S_L > S_{\text{allow}}$$

$$310.40 \text{ Mpa} > 183.40 \text{ Mpa} \rightarrow \text{"Failed"}$$

Calculation of Allowable Stress:

For A358-304, from ASME B31.3

$$S_Y = 30 \text{ ksi} = 206.84 \text{ Mpa}$$

$$S_U = 75 \text{ ksi} = 517.10 \text{ Mpa}$$

$$S_H = S_Y / 1.5 = 206.84 \text{ Mpa} / 1.5$$

$$S_H = 137.9 \text{ Mpa}$$

$$S_H = S_U / 3.0 = 517.10 \text{ mpa} / 3.0$$

$$S_H = 172.35 \text{ Mpa}$$

Use the lower value,

$$S_H = 137.9 \text{ Mpa}$$

$$S_{\text{allow}} = 1.33 \times 137.9 \text{ Mpa}$$

$$S_{\text{allow}} = 183.40 \text{ Mpa}$$

PIPING FLEXIBILITY AND STRESS ANALYSIS

DISPLACEMENT STRESS RANGE

Calculation of Displacement Stress:

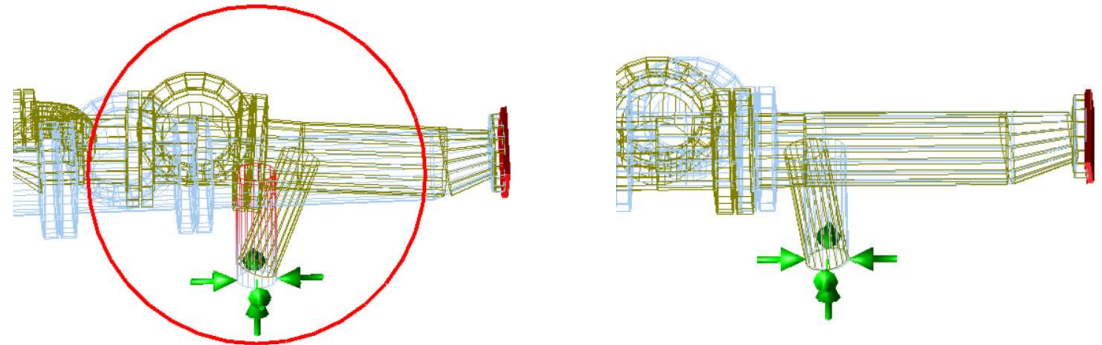
The location of highest sustained stress is at node 40.

Pipe Diameter = 355.6 mm

$S_a = 0$

Bending Stress, **S_b** = 422.2 Mpa (by CII)

Torsional Stress, **S_t** = 0.00 Mpa (by CII)



PIPING FLEXIBILITY AND STRESS ANALYSIS

DISPLACEMENT STRESS RANGE

Calculation of Displacement Stress:

$$S_E = \sqrt{[S_a + S_b]^2 + (2S_t)^2}$$

$$S_E = \sqrt{[0 + 422.2]^2 + (2 \times 0.0)^2}$$

$$S_E = 422.2 \text{ Mpa}$$

Evaluation:

$$S_L > S_{\text{allow}}$$

$$422.2 \text{ Mpa} > 206.85 \text{ Mpa} \rightarrow \text{"Failed"}$$

Calculation of Allowable Stress:

For A358-304, from ASME B31.3

$$S_Y = 30 \text{ ksi} = 206.84 \text{ Mpa}$$

$$S_U = 75 \text{ ksi} = 517.10 \text{ Mpa}$$

$$S_H = S_Y / 1.5 = 206.84 \text{ Mpa} / 1.5$$

We already know that the lower value is 137.9 Mpa

$$S_H = S_C = 137.9 \text{ Mpa}$$

$$S_{\text{allow}} = f (1.25S_C + 0.25S_H)$$

$$S_{\text{allow}} = f [(1.25 \times 137.9) + (0.25 \times 137.9)]$$

$$S_{\text{allow}} = 206.85 \text{ Mpa}$$

PIPING FLEXIBILITY AND STRESS ANALYSIS

FLEXIBILITY ANALYSIS

Flexibility Analysis:

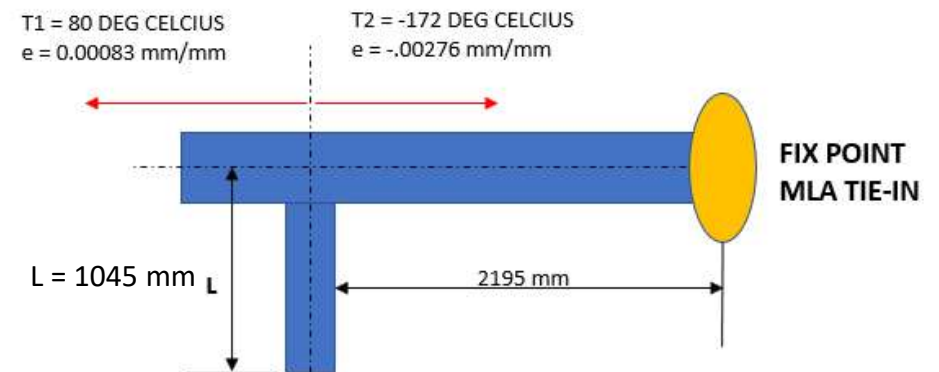
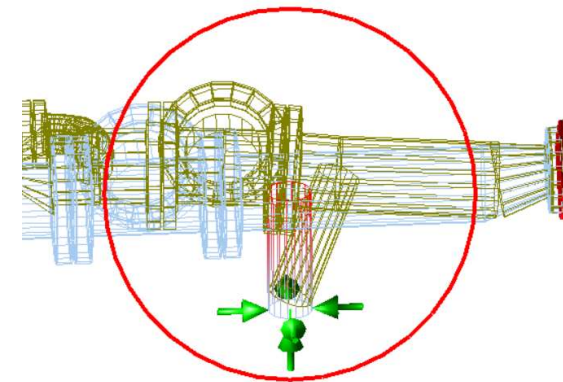
The length of the deformed leg can be estimated using guided cantilever method.

$$L_{\text{req}} = \sqrt{\frac{3ED\Delta}{S}}$$

Estimating Leg Length:

$$L_{\text{req}} = \sqrt{\frac{3 \times 195,490 \text{ Mpa} \times 355 \text{ mm} \times 6.0582 \text{ mm}}{206.85 \text{ Mpa}}}$$

$$L_{\text{req}} = 2,469.3 \text{ mm}$$

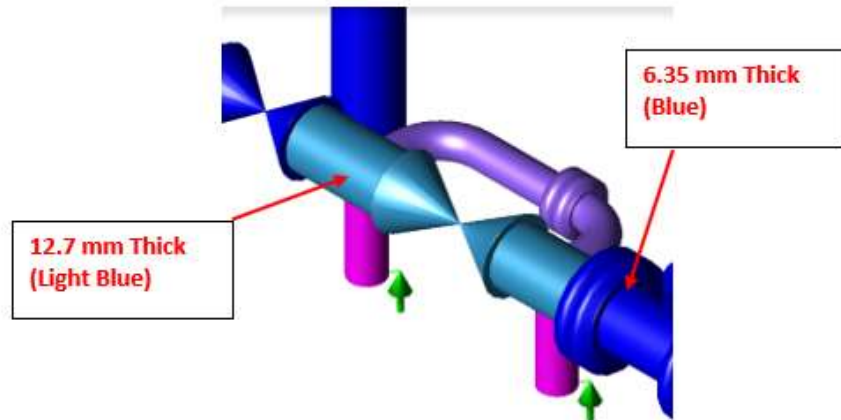


EXPANSION @ T1 = 2195 mm x 0.00083 mm/mm = 1.821 mm
 CONTRACTION @ T2 = 2195 mm x -0.00276 mm/mm = -6.0582 mm

PIPING FLEXIBILITY AND STRESS ANALYSIS: MODIFICATION, ADJUSTMENT & RECOMMENDATIONS

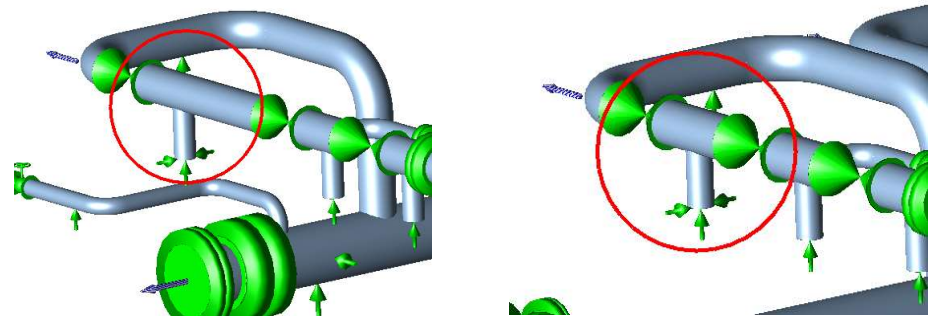
For High Sustained Stress:

1. Increased pipe thickness.
2. Add support upstream of the valve.



For High Occasional Stress:

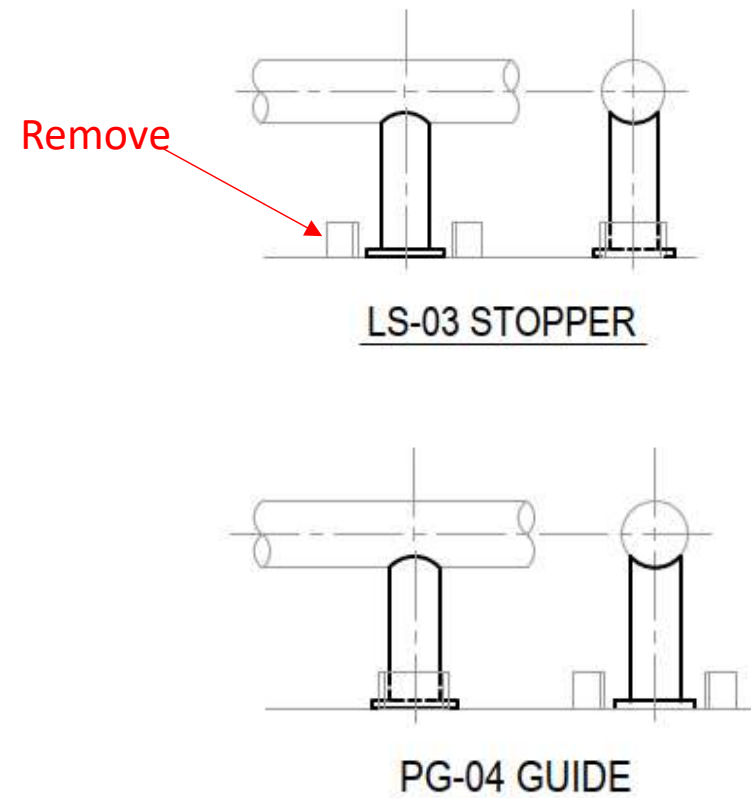
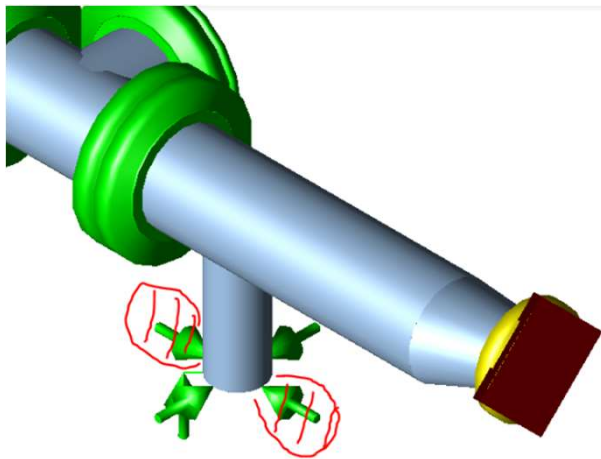
1. Same modification for the High sustained stress, but with;
2. Additional guides on some supports.



PIPING FLEXIBILITY AND STRESS ANALYSIS: MODIFICATION, ADJUSTMENT & RECOMMENDATIONS

For High Displacement Stress:

1. Remove the pipe stopper



PIPING FLEXIBILITY AND STRESS ANALYSIS: RE-PROCESSING ANALYSIS

Re-Calculation of Sustained Stress:

The location of highest sustained stress is at tee, node **610**.

Pipe Diameter = 406.4 mm

$S_a = SLP + F/A$; $F/A \rightarrow$ can be neglected

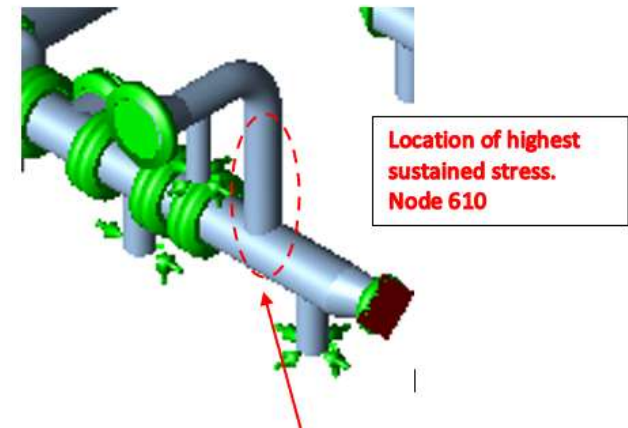
$SLP = PD/4t$

$SLP = [1.8\text{Mpa} \times 406.4\text{mm}] / [4 \times 4.77\text{ mm}]$

SLP = 38.3 Mpa

Bending Stress, **Sb** = 72.2 Mpa (by CII)

Torsional Stress, **St** = -0.50 Mpa (by CII)



PIPING FLEXIBILITY AND STRESS ANALYSIS: RE-PROCESSING ANALYSIS

Re-Calculation of Sustained Stress:

Highest Sustained Stress at Node 610;

$$S_L = \sqrt{[S_a + S_b]^2 + (2S_t)^2}$$

$$S_L = \sqrt{[38.3 + 72.2]^2 + (2 \times 0.50)^2}$$

$$S_L = 110.5 \text{ Mpa}$$

Evaluation:

$$S_L < S_H$$

$$110.5 \text{ Mpa} < 137.9 \text{ Mpa} \rightarrow \text{"Pass"}$$

Re-checking the stresses at node 450.

Longitudinal Pressure Stress, SLP

$$SLP = PD/4t$$

$$SLP = (1.8 \text{ Mpa} \times 609.6 \text{ mm}) / (4 \times 12.7 \text{ mm})$$

$$SLP = 21.6 \text{ Mpa}$$

Bending Stress, Sb = 2.5 Mpa (by CII)

Torsional Stress, St = -0.10 Mpa (by CII)

$$S_L = \sqrt{[21.6 + 2.5]^2 + (2 \times 0.10)^2}$$

$$S_L = 24.10 \text{ Mpa}$$

PIPING FLEXIBILITY AND STRESS ANALYSIS: RE-PROCESSING ANALYSIS

Re-Calculation of Occasional Stress:

The location of highest sustained stress is at node **1520**.

Pipe Diameter = 273 mm

$S_a = SLP + F/A$; $F/A \rightarrow$ can be neglected

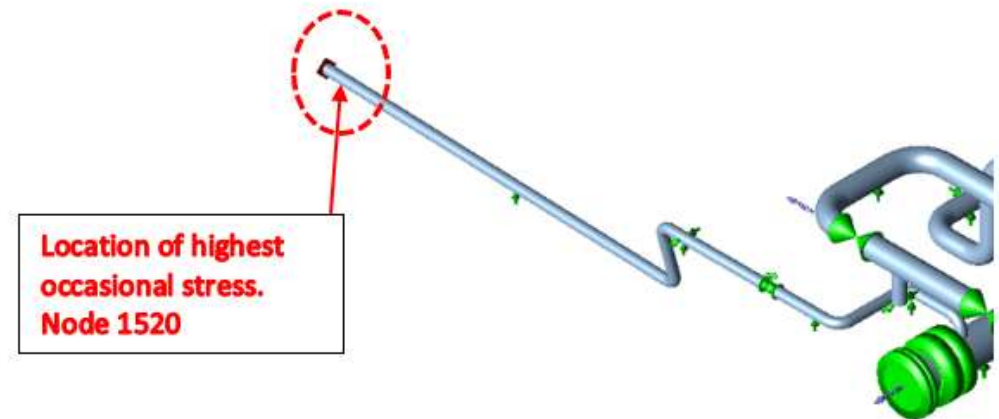
$SLP = PD/4t$

$SLP = [1.8\text{Mpa} \times 273\text{mm}] / [4 \times 4.2 \text{ mm}]$

SLP = 29.3 Mpa

Bending Stress, **S_b** = 129.9 Mpa (by CII)

Torsional Stress, **S_t** = -14.6 Mpa (by CII)



PIPING FLEXIBILITY AND STRESS ANALYSIS: RE-PROCESSING ANALYSIS

Re-Calculation of Occasional Stress:

$$S_{Loc} = \sqrt{[S_a + S_b]^2 + (2S_t)^2}$$

$$S_{Loc} = \sqrt{[29.3 + 129.9]^2 + (2 \times 14.6)^2}$$

$$S_{Loc} = \mathbf{161.8 \text{ Mpa}}$$

Evaluation:

$$S_{Loc} < 1.33S_H$$

$$161.8 \text{ Mpa} < 183.4 \text{ Mpa} \rightarrow \text{"Pass"}$$

Re-checking the stresses at node 450.

Longitudinal Pressure Stress, SLP

$$SLP = PD/4t$$

$$SLP = (1.8 \text{ Mpa} \times 609.6 \text{ mm}) / (4 \times 12.7 \text{ mm})$$

$$SLP = \mathbf{21.6 \text{ Mpa}}$$

Bending Stress, Sb = **39.6** Mpa (by CII)

Torsional Stress, St = -0.10 Mpa (by CII)

$$S_{Loc} = \sqrt{[21.6 + 39.6]^2 + (2 \times 0.10)^2}$$

$$S_{Loc} = 61.20 \text{ Mpa}$$

PIPING FLEXIBILITY AND STRESS ANALYSIS: RE-PROCESSING ANALYSIS

Re-Calculation of Displacement Stress:

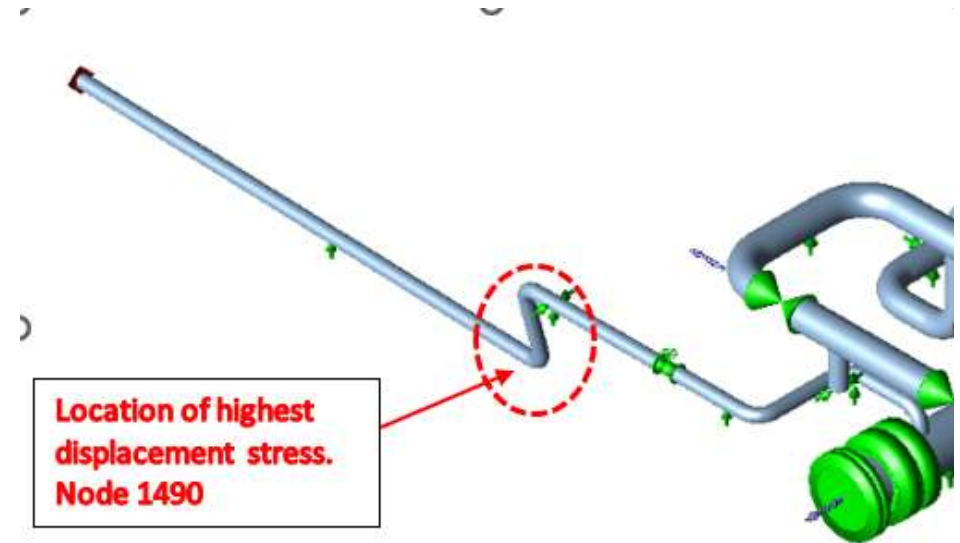
The location of highest sustained stress is at node **1490**.

Pipe Diameter = 273 mm

$S_a = 0$

Bending Stress, **S_b** = 232.2 Mpa (by CII)

Torsional Stress, **S_t** = -0.90 Mpa (by CII)



PIPING FLEXIBILITY AND STRESS ANALYSIS: RE-PROCESSING ANALYSIS

Re-Calculation of Displacement Stress:

$$S_E = \sqrt{[S_a + S_b]^2 + (2S_t)^2}$$

$$S_E = \sqrt{[0 + 232.2]^2 + (2 \times 0.9)^2}$$

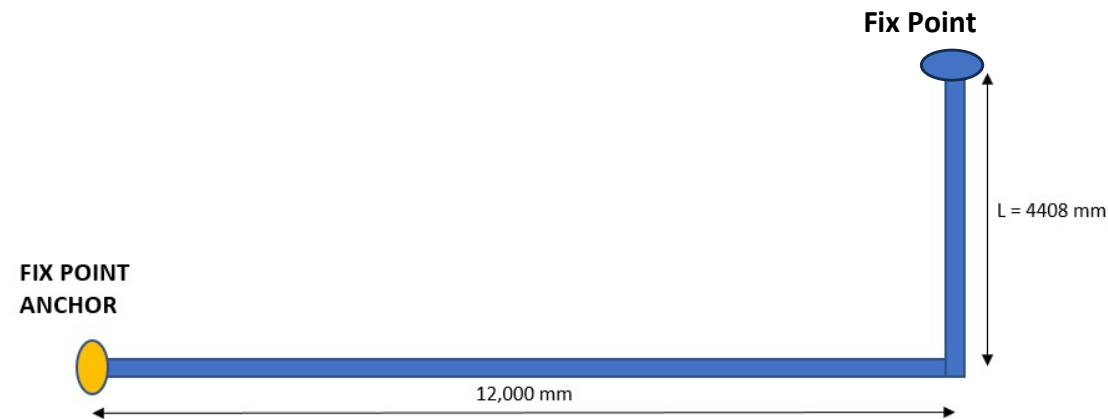
$$S_E = 232.2 \text{ Mpa}$$

Evaluation:

$$S_L > S_{\text{allow}}$$

$$232.2 \text{ Mpa} > 206.85 \text{ Mpa} \rightarrow \text{"Failed"}$$

Flexibility Analysis:



EXPANSION @ T1 = 12000 mm x 0.00083 mm/mm = 9.96 mm
CONTRACTION @ T2 = 12000 mm x -0.00276 mm/mm = -33.12 mm

$$L_{\text{req}} = \sqrt{\frac{3ED\Delta}{S}}$$

PIPING FLEXIBILITY AND STRESS ANALYSIS: RE-PROCESSING ANALYSIS

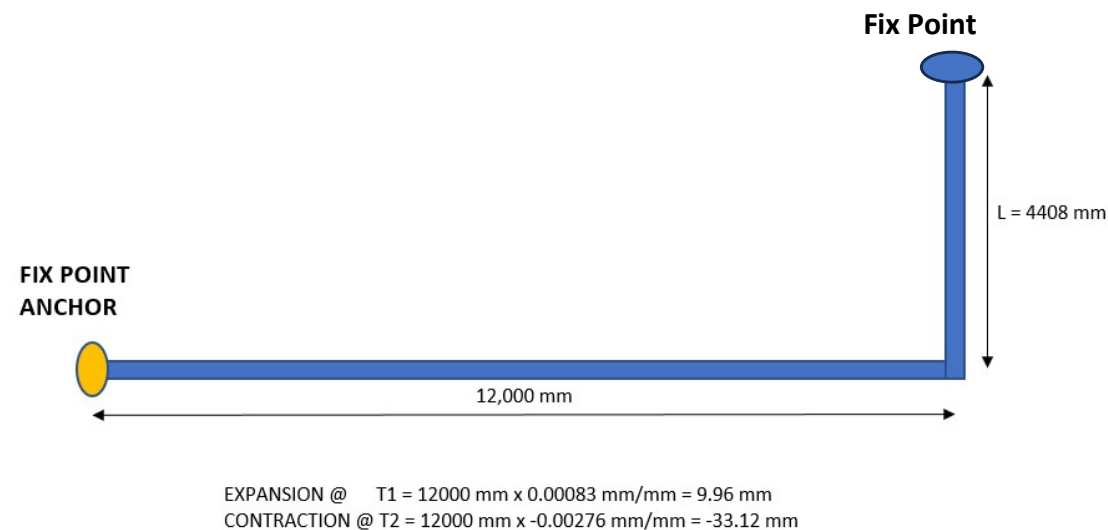
Flexibility Analysis:

Estimating Leg Length:

$$L_{\text{req}} = \sqrt{\frac{3ED\Delta}{S}}$$

$$L_{\text{req}} = \sqrt{\frac{3 \times 195,490 \text{ Mpa} \times 273 \text{ mm} \times 43.08 \text{ mm}}{206.85 \text{ Mpa}}}$$

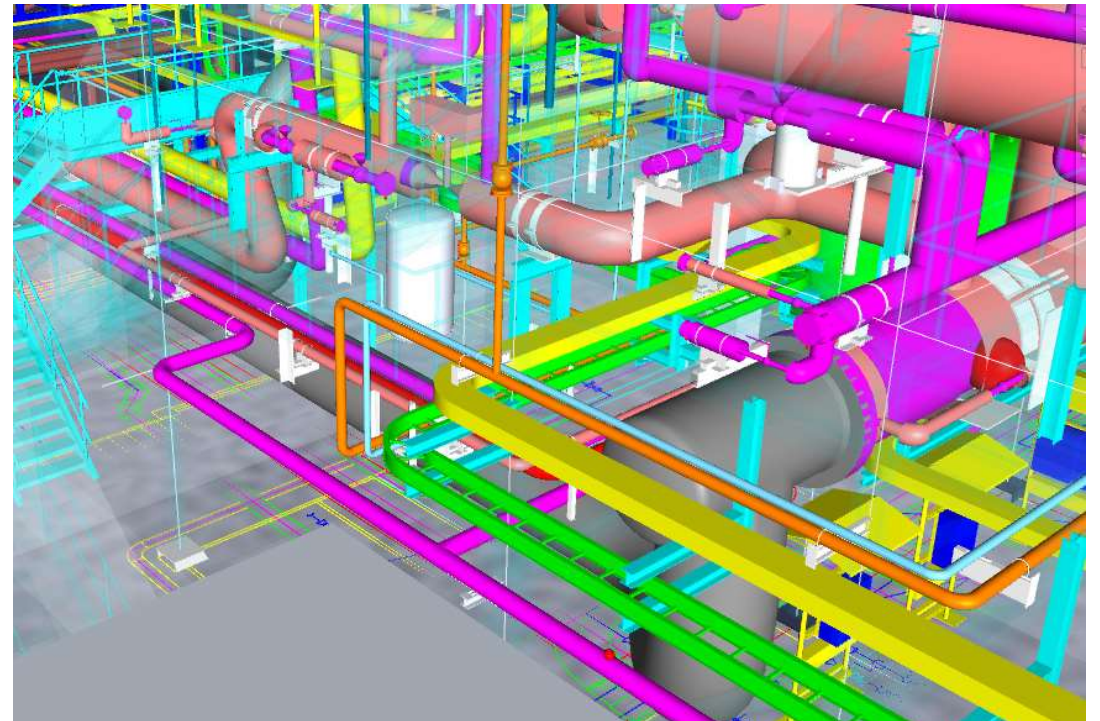
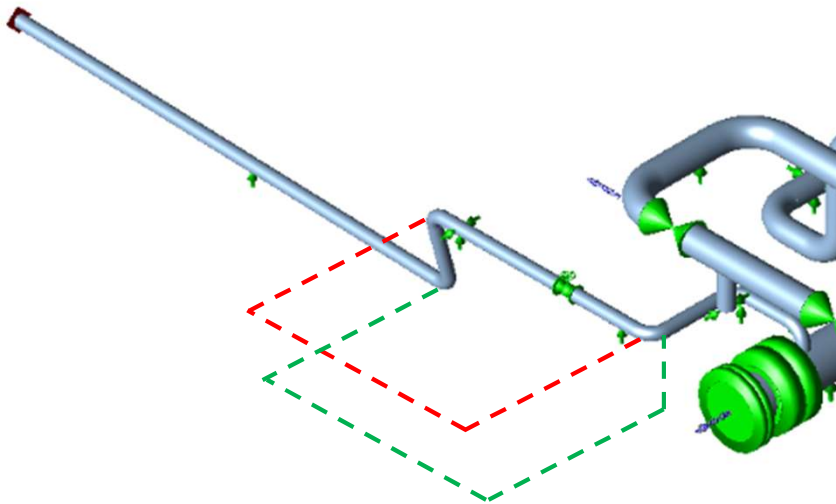
$$L_{\text{req}} = 5,774.5 \text{ mm}$$



$$L_{\text{req}} = \sqrt{\frac{3ED\Delta}{S}}$$

PIPING FLEXIBILITY AND STRESS ANALYSIS: INVESTIGATE POSSIBLE ROUTE IN 3D MODEL

Possible Route:



PIPING FLEXIBILITY AND STRESS ANALYSIS

ENGINEERING JUDGMENT:

Engineering Judgment:

Use the sustained load margin by adding it to the displacement stress range allowable.

$$S_E = \sqrt{[S_a + S_b]^2 + (2S_t)^2}$$

$$S_E = \sqrt{[0 + 232.2]^2 + (2 \times 0.9)^2}$$

$$S_E = 232.2 \text{ Mpa}$$

Evaluation:

$$S_L < S_{\text{allow}}$$

$$232.2 \text{ Mpa} < 311.75 \text{ Mpa} \rightarrow \text{"Pass"}$$

Considering Sustained Load Margin:

The sustained stress SL at node 1490

SL = 33.0 Mpa (by software)

$$S_{\text{allow}} = f [(1.25S_C + 0.25S_H) + (S_H - S_L)]$$

$$S_{\text{allow}} = f [(1.25 \times 137.9) + (0.25 \times 137.9) + (137.9 - 33.0)]$$

$$S_{\text{allow}} = 311.75 \text{ Mpa}$$

DESIGN OF LNG BUFFER TANK

ASME BPVC Section VIII, Division 1



DESIGN OF LNG BUFFER TANK

The minimum required thickness of shell under pressure shall not less than that computed by the following formulas, *ASME BPVC Section VIII Division 1, UG-27(c)*.

- Circumferential Stress (Longitudinal Joints). When $t < \frac{1}{2} R$ and $P < 0.385SE$

$$t_1 = \frac{PR}{SE - 0.6P}$$

- Longitudinal Stress (Circumferential Joint). When $t < \frac{1}{2} R$ and $P < 1.25SE$

$$t_2 = \frac{PR}{2SE + 0.4P}$$

DESIGN OF LNG BUFFER TANK

VESSEL DESIGN DATA:

MATERIAL	SA-353 UNS K81340
ULTIMATE STRENGTH, S_u	690 MPa
YIELD STRENGTH, S_y	515 MPa
DESIGN CONDITION	1.8 MPa @ 80°C
INSIDE DIAMETER, D_i	3800 mm
CORROSION ALLOWANCE, CA	0.0 mm
WELD JOINT EFFICIENCY, E	1.0
MINIMUM DESIGN METAL TEMP.	-162°C

ALLOWABLE STRESS:

Choose the lower between the two;

$$S_1 = 690 \text{ Mpa} / 3.5 = \underline{197.14 \text{ Mpa}}$$

$$S_2 = 515 \text{ Mpa} / 1.5 = 343.33 \text{ Mpa}$$

DESIGN OF LNG BUFFER TANK

Calculation of Shell Thickness:

Determine the inside radius, R

$$R = (D_i + 2CA) / 2 = [3800 \text{ mm} + (2 \times 0 \text{ mm})] / 2$$

$$R = 1900 \text{ mm}$$

Determine the Static Liquid Pressure, $P_s = \rho gh$

Assume that the pressure vessel is full and the maximum level of the fluid inside the horizontal vessel is 3800 mm.

$$P_s = 1000 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times 3.80 \text{ m}$$

$$P_s = 37,278 \text{ Pascal} = 0.0373 \text{ Mpa}$$

Determine the Design Pressure

$$P = 1.80 \text{ Mpa} + 0.0373 \text{ Mpa}$$

$$P = 1.8373 \text{ Mpa}$$

Determine the Thickness due to Circumferential Stress t_1 ;

$$t_1 = \frac{PR}{SE - 0.6P}$$

$$t_1 = \frac{1.8373 \times 1900}{(197.14 \times 1.0) - (0.60 \times 1.8373)}$$

$$t_1 = 17.81 \text{ mm} \rightarrow \text{required thickness}$$

DESIGN OF LNG BUFFER TANK

Calculation of Shell Thickness:

Design thickness = Required t + CA

Design thickness = 17.81 mm + 0 mm

Design thickness = 18 mm

Checking the condition of ASME Section VIII

$T \leq 0.50R$

18 mm $\leq 0.5 \times 1900$

18 mm ≤ 950 mm $\rightarrow$ "ok"

$P \leq 0.385SE$

1.8373 Mpa $\leq 0.385 \times 197.14 \times 1.0$

1.8373 Mpa ≤ 75.9 Mpa $\rightarrow$ "ok"

Determine the Thickness due to Longitudinal Stress t_2 ;

$$t_2 = \frac{PR}{2SE + 0.4P}$$

$$t_2 = \frac{1.8373 \times 1900}{(2 \times 197.14 \times 1.0) + (0.4 \times 1.8373)}$$

$t_2 = 8.84$ mm $\rightarrow$ Required thickness

Design thickness = Required t + CA

$t_2 = 8.84$ mm + 0 mm

$t_2 = 8.84$ mm = 9 mm

DESIGN OF LNG BUFFER TANK

Calculation of Shell Thickness:

The shell thickness shall be the greater between the computed design thickness above;

Use design thickness = 18 mm

DESIGN OF LNG BUFFER TANK

Calculation of Head Thickness:

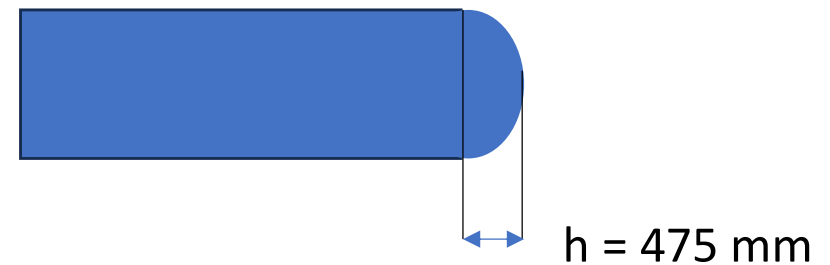
The minimum required thickness at the thinnest point after forming of ellipsoidal head shall be computed by the following equation.

$$t_e = \frac{PDK}{2SE - 0.2P}$$

Solving for K factor

$$K = \frac{1}{6} \left[2 + \left(\frac{D}{2h} \right)^2 \right]$$

Determine the h dimension



$$K = \frac{1}{6} \left[2 + \left(\frac{1900}{2 \times 475} \right)^2 \right]$$

$$K = 1.0$$

DESIGN OF LNG BUFFER TANK

Calculation of Head Thickness:

$$t_e = \frac{PDK}{2SE - 0.2P}$$

$$t_e = \frac{1.8373 \times 3800 \times 1.0}{(2 \times 197.14 \times 1.0) - (0.2 \times 1.8373)}$$

$$t_e = 17.724 \text{ mm}$$

Design thickness = t_e + CA

Design thickness = 17.724 + 0 mm

Design thickness = 17.724 mm

= say 18 mm

Calculation of Tank Length:

Volume of tank = 400 m³

Inside Diameter = 1900 mm

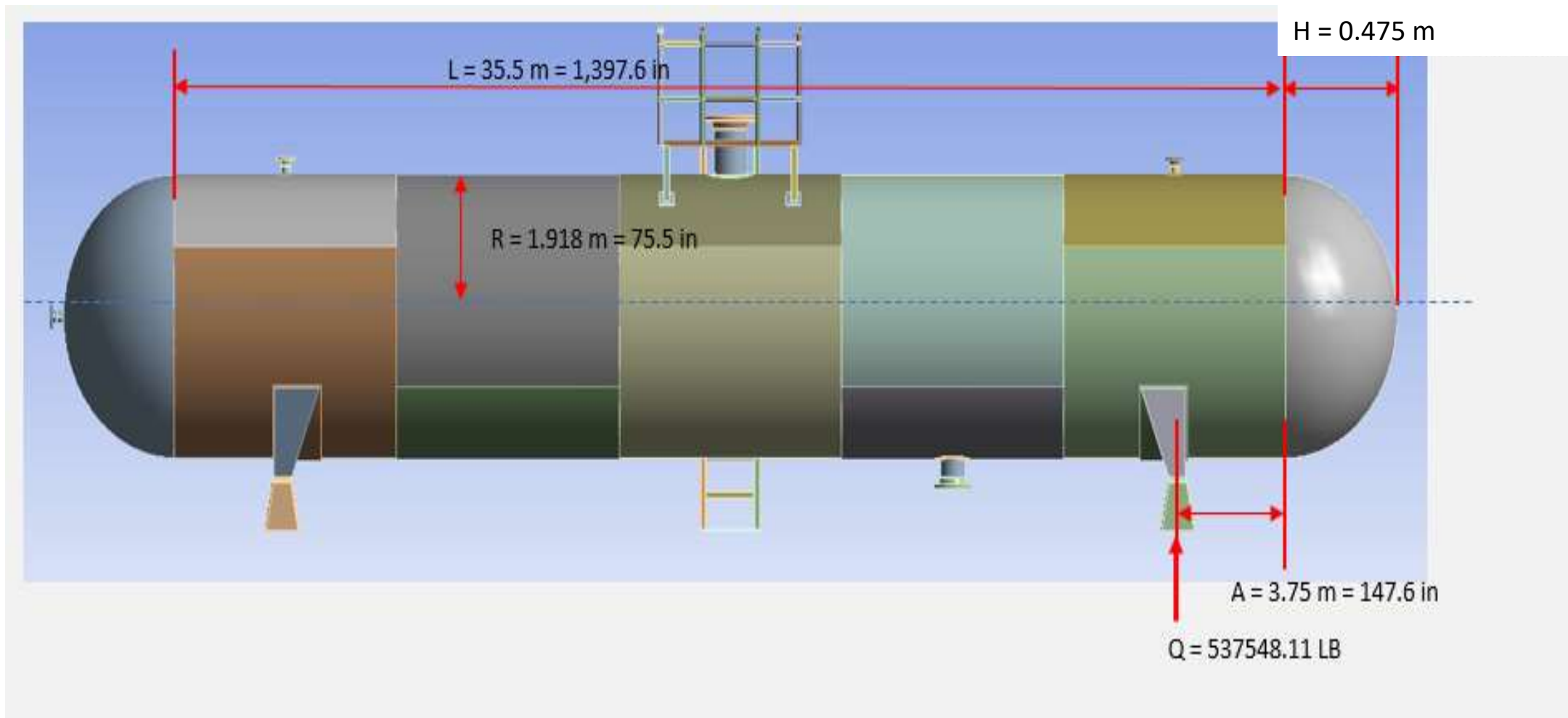
$$\text{Volume} = \frac{\pi}{4} D^2 L$$

$$\text{Length} = \frac{4 \times 400}{\pi \times 1900^2}$$

Length = 35.26 m

= say 35.50 m

DESIGN OF LNG BUFFER TANK



HYDROSTATIC TEST PRESSURE OF PIPING.

- The test fluid shall be water and according to ASME B31.3, para. 345.4.1. The test pressure at every point in a metallic piping shall not be less than 1.5 times the design pressure or when the design temperature is greater than the test temperature, the minimum test pressure shall be calculated using
- $P_t = 1.5 P \times \left(\frac{S_T}{S}\right)$
- $P_t = 1.5 \times 1.8 \times (137.9/139.7)$
- $P_t = 2.70 \text{ Mpa}$

HYDROSTATIC TEST PRESSURE OF LNG TANK.

- The test fluid shall be water and the test are done after the fabrication is completed. The test pressure shall be calculated as per ASME section VIII Division 1, UG-99.

- $P_T = 1.3 \text{ MAWP} \times \text{LSR}$

Where: MAWP = maximum allowable working pressure

LSR = lowest stress ratio of material at design temperature versus test temperature.

- $P_T = 1.3 \times 1.82 \text{ MPa} \times 1.0$

- $P_T = 2.37 \text{ MPa}$

COST OF FAILURE

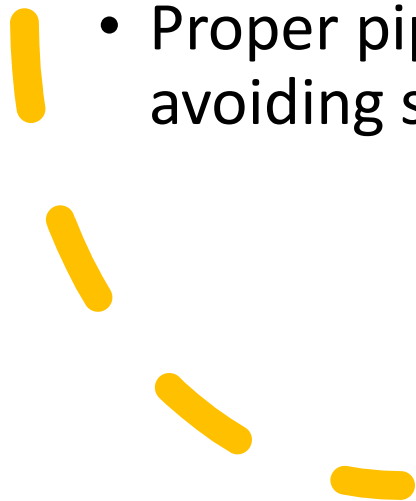
- Engineers design facilities to prevent equipment failure.
- ASME provides guidelines for designing piping systems and pressure vessels for safe operation.
- Equipment failure leads to production loss, operation shutdown, and various costs.
- Costs include repair, maintenance, materials, labor, energy losses, and inefficiencies.
- Complete plant shutdown results in lost profits, wasted costs, start-up losses, and repair expenses.

MINIMIZING PIPING MATERIAL AND SUPPORTS DURING DESIGN STAGE.

- Designers aim to consider piping support span and flexibilities.
- Some designers lack experience and produce overly conservative or liberal designs.
- Checking involves minimizing pipe fittings and supports based on calculated span and stress results.
- Unnecessary pipe supports may be removed based on calculations and engineering experience.
- Materials like elbows are checked for flexibility needs; excessive flexibility can be eliminated.
- Reducing costs is considered by optimizing design elements like fittings and supports.
- Economic aspects are addressed during the design stage.

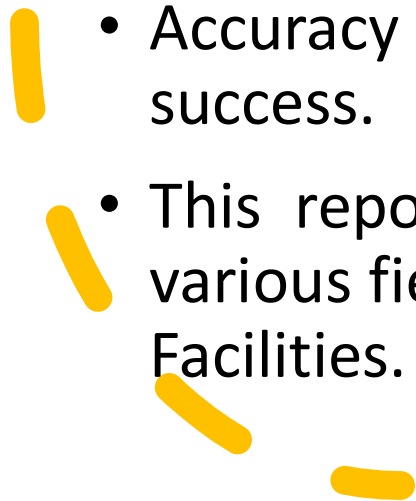


RECOMMENDATIONS

- Codes and standards are crucial in Oil and Gas, Energy, and Power projects to enhance safety.
 - Data and structural information, including seismic and wind data, are essential in design.
 - Pipe wall thickness calculations should consider both internal and external design pressures.
 - Proper pipe support placement is critical for low-stress piping and avoiding sagging.
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RECOMMENDATIONS

- A 25% increase in equivalent length is recommended for future bends and rerouting.
 - ASME code requirements emphasize the need for piping stress analysis, performed by specialized mechanical engineers.
 - Finite Element Analysis (FEA) software like CAESAR II and PV Elite is valuable for stress and pressure vessel analysis.
 - Accuracy and speed in calculations are essential for industry success.
 - This report serves as a reference for mechanical engineers in various fields, including Oil & Gas, Energy & Power, and Advanced Facilities.
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REFERENCES

- ✓ ASME B31.3 PROCESS PIPING CODE – 2020
- ✓ ASME BPVC SECTION VIII, DIVISION 1 – 2019
- ✓ ASME BPVC SECTION II, PART D – 2019
- ✓ ASCE 7-10 MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES
- ✓ DESIGN OF PIPING SYSTEMS, KELLOG COMPANY
- ✓ PIPING SYSTEMS MANUAL, BRIAN SILOWASH
- ✓ PIPING AND PIPELINE CALCULATIONS MANUAL, PHILIP ELLENBERGER
- ✓ PIPE STRESS ENGINEERING, PENG

THANK YOU.